



In partnership with the University of New Mexico, Bosque School, and Sevilleta LTER

BOSQUE ECOSYSTEM MONITORING PROGRAM (BEMP)
SITE MONITORING REPORT FOR 2022

2022 ANNUAL SITE MONITORING TECHNICAL REPORT

Submitted

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1 Introduction

Objective

To collect and analyze abiotic and biotic data at BEMP sites in the Middle Rio Grande Bosque while involving K-12 and university students in learning about and monitoring this ecosystem. All data and reports are available on the BEMP website, www.BEMP.org

Scope of Work

The Bosque Ecosystem Monitoring Program (BEMP) combines long-term ecological research with community outreach by involving K-12 teachers and their students in monitoring key indicators of structural and functional change in the Middle Rio Grande riparian forest, or “bosque.” In 1996, BEMP began as a program of the University of New Mexico’s Department of Biology (under NSF Grant No. DEB-9420510, Amendment No. 004) and quickly became a collaboration between the University of New Mexico and Bosque School in Albuquerque, with fewer than 200 participants in its first year. Before the COVID-19 pandemic, BEMP was averaging approximately 9,000 participants annually. The BEMP outdoor education experience builds science skills, educates the community about the bosque ecosystem, and helps create a constituency for stewardship of the bosque. BEMP findings derived from student-gathered data are used by government agencies to inform multi-million dollar river and management decisions that impact the riparian corridor.

The 2022 reporting period covers 33 BEMP sites along 250 miles of the Rio Grande, including 32 sites within the Middle Rio Grande (Figure 1.1). Through the stakeholder driven and strategic location of these sites, BEMP studies the ecological drivers and effects of fire, flooding, climate change, and human alteration on the bosque ecosystem. Two-thirds of BEMP sites were installed at the request of natural resource managers to monitor the long-term ecological impacts of restoration projects such as mechanical clearing, wood chipping, and bank-lowering. The other third were installed by BEMP staff to facilitate research opportunities or at the request of schools or other partners.

BEMP monitors biotic and abiotic variables at our research sites. Our abiotic datasets include: depth to groundwater; water level in ditches and drains; precipitation; above- and

below-ground temperature; and water quality in the Rio Grande. Our biotic datasets include litterfall; vegetation cover; surface-active arthropod richness and abundance; and tamarisk leaf beetle distribution, abundance and impact.

The Valle de Oro BEMP site was under construction from September 2019 through August 2022 and was reinstalled in September 2022. The Pueblo of Santo Domingo was closed to all visitors during the COVID-19 restrictions and the site remained off-line for most of 2020 through the summer of 2022. The Pueblo opened up to visitors and BEMP monitoring was resumed in June of 2022.

BEMP hosted two events during 2022 to present new data, visualizations, and analyses: the Crawford Symposium and the Luquillo-Sevilleta Virtual Spanish Symposium. Both these events feature student presentations. BEMP staff and students present BEMP data to managers, professionals, and students several times throughout the year depending on conference availability. In 2022, BEMP data were shared at conferences and workshops including the Sevilleta All Hands Meeting. BEMP staff participated and consulted in a post-burn tour following the Big Hole fire in Belen at Whitfield Wildlife Conservation Area. BEMP staff co-hosted a tour of the Albuquerque fire with the City of Albuquerque and Western EcoSystems Technology, Inc. in June 2022. There were 38 participants from various agencies. BEMP also participated in the post-burn Task Force to address concerns and plans going forward.

Timing of Data Collection

Depth to groundwater, water level in nearby ditches and/or drains, precipitation, and litterfall are collected monthly, during the week of the third Tuesday of each month. Surface-active arthropods are collected three times each year, in the spring, summer, and fall. Vegetation cover is surveyed once each year in August-September. Tamarisk leaf beetle monitoring is conducted during the week of monthly monitoring from May-August, with some sites collected through September. All other datasets are collected as funding permits.

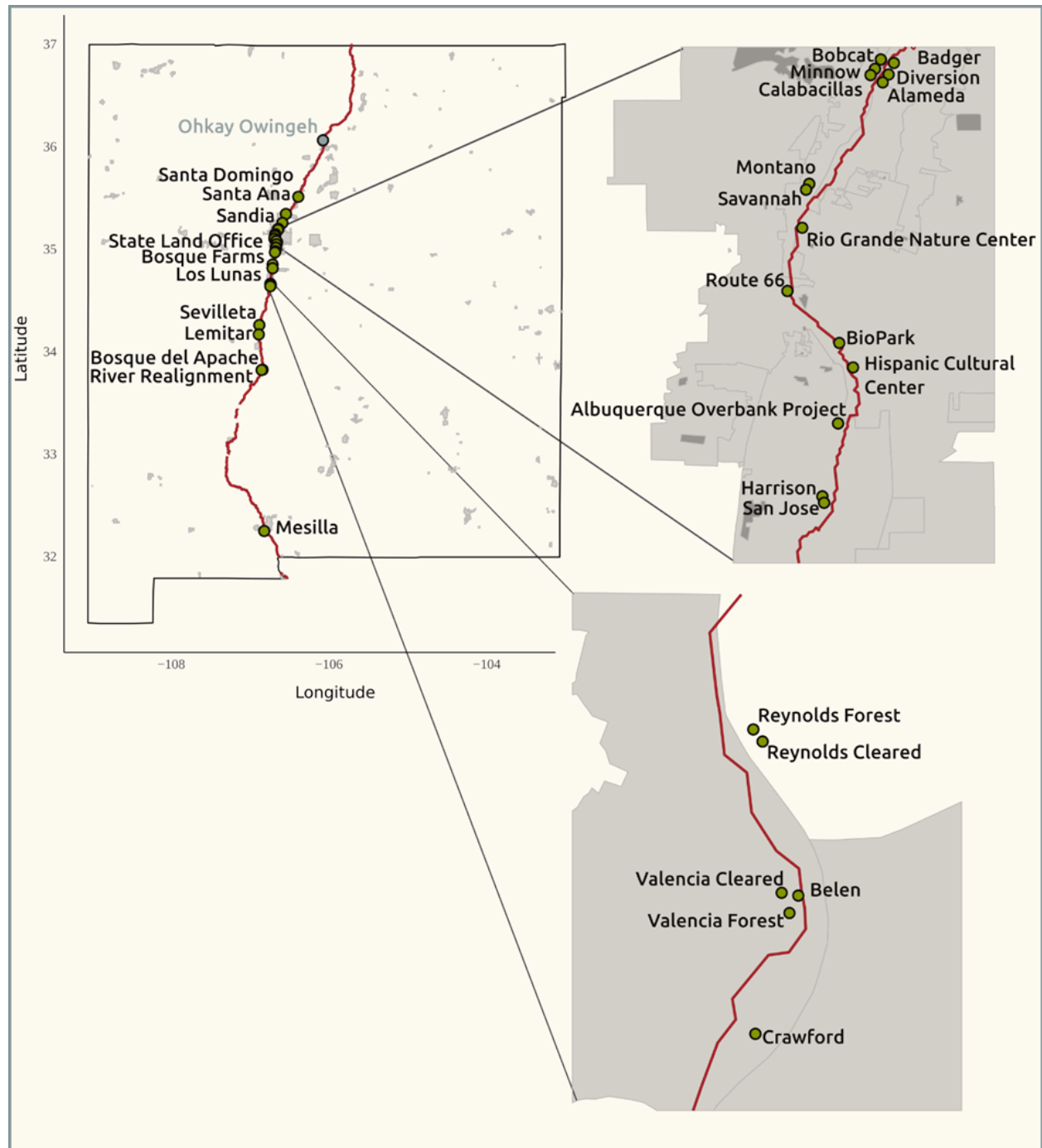


Figure 1.1. Map of 33 active BEMP sites along the Rio Grande; Ohkay Owingeh is no longer an active site.

Table 1.1. BEMP sites and locations along the Rio Grande by Reach, listed from north to south.

* denotes inactive site (either no longer active or temporarily inactive)

Site number	Site name	Latitude	Longitude	Reach
9	Ohkay Owingeh*	36.0618	-106.0761	Cochiti
24	Santo Domingo	35.50989	-106.3896	Cochiti
5	Santa Ana	35.34284	-106.5458	Angostura
32	Sandia	35.255	-106.5907	Angostura
22	Bobcat	35.19705633	-106.6439494	Angostura
21	Badger	35.1956	-106.6402	Angostura
12	Minnow	35.19315094	-106.646915	Angostura
10	Diversion	35.1908	-106.6429	Angostura
11	Calabacillas	35.19056822	-106.6491626	Angostura
1	Alameda	35.1875	-106.6459	Angostura
17	Montano	35.14528819	-106.6803699	Angostura
6	Savannah	35.14285294	-106.6819814	Angostura
2	Rio Grande Nature Center (RGNC)	35.127	-106.6854	Angostura
20	Route 66	35.1006408	-106.6914783	Angostura
23	BioPark	35.079	-106.668	Angostura
8	Hispanic Cultural Center (HCC)	35.06881267	-106.6580575	Angostura
29	Albuquerque Overbank Project (AOP)	35.04546	-106.6657	Angostura

13	Harrison	35.01505603	-106.6736953	Angostura
31	San Jose	35.012375	-106.6728	Angostura
28	Valle de Oro	34.97895	-106.6801	Angostura
30	State Land Office (SLO)	34.96785	-106.6856	Angostura
27	Bosque Farm	34.848851	-106.714722	Isleta
3	Los Lunas	34.81236936	-106.714458	Isleta
19	Reynolds Forest	34.66054583	-106.7429525	Isleta
18	Reynolds Cleared	34.65966431	-106.7421328	Isleta
15	Valencia Cleared	34.64863444	-106.7391728	Isleta
4	Belen	34.6484315	-106.7377022	Isleta
16	Valencia Forest	34.64716225	-106.738482	Isleta
25	Crawford	34.63835	-106.74277	Isleta
14	Sevilleta	34.25834233	-106.8831845	San Acacia
7	Lemitar	34.16703188	-106.8899486	San Acacia
34	River Realignment	33.8227	-106.8419	San Acacia
33	Bosque del Apache (BDA)	33.8197	-106.8539	San Acacia
26	Mesilla	32.248328	-106.821014	South of San Marcial

Site Update

In April 2022, the Big Hole fire in Valencia County burned the northern half of the Reynolds Cleared site (site 18) and the entire Reynolds Forest site (site 19). Site repair took place the following month. In May 2022, a bosque fire south of Montano bridge burned approximately 30 acres of forest, including the southern side of the Savannah site (site 6). Site repairs were carried out after the bosque reopened. In June 2022, after heavy monsoon rains, the Sevilleta site (site 14) flooded with evidence of water reaching 73 cm above ground level at some locations.

2 Importance of long-term data, community science, and education outreach

BEMP's mission is community science, education, and stewardship: equitable and inclusive hands-on student research essential to the management of the Rio Grande ecosystem.

The long-term data generated by BEMP have been used in informing predictive models, assessing restoration projects, tracking shifts in native and exotic vegetation, understanding bosque response to different ecosystem drivers (e.g., fire, flooding, clearing, impacts of climate change, introduction of biocontrol species), determining what ecotones are present, and how to gracefully transition former riparian areas in sustainable semi-arid ecosystems. Long-term monitoring of these sites is critical for understanding how the ecosystem responds to land management strategies and climate variability under rapidly changing means and variances. Long term data is necessary for effectively applying adaptive management and developing best practices strategies.

BEMP involves community members and students of all ages, from pre-K through high school, college, graduate, to life-long learners volunteering in the program. Our primary focus is on elementary, middle school, and high school students that participate in monthly fieldwork (long term educational engagement) to collect groundwater, precipitation, and litterfall data, as well as going out once or twice each year to participate in monitoring arthropods. Students have opportunities to participate in other data collections, including water quality, tamarisk leaf beetle, and monitoring fuel load. BEMP involves UNM undergraduate and graduate students in a semester-long internship experience through an upper division biology BEMP course, BIOL 408/508, where they learn about the bosque ecosystem, develop independent projects applying BEMP data, work with K-12 students and teachers, and play an integral role in regular field and lab work. The work of K-12 students in the field is facilitated and quality controlled by BEMP staff as well as the UNM interns. Having now played a role in our community for a couple of decades, we are starting to see the long-term impacts of our programming. Each year, there are a few UNM students in the BEMP course that had previously participated in BEMP as elementary, middle, and/or high school students. These students are often reconnected to their former schools and sites. BEMP has been part of a meaningful story for many students and community members. BEMP helps students connect with their local landscape, learn science through hands-on research, and communicate or present their understanding through math, writing, art, and other forms of expression. Several former BEMP students now have jobs with our partner agencies, including Bernalillo County

Water Utility Authority, City of Albuquerque Open Space, Middle Rio Grande Conservancy District, and New Mexico Interstate Stream Commission.

3 Temperature

During the 2021-2022 reporting period, we collected data from LogMaster temperature loggers at nine BEMP sites. Three loggers were initially installed at each selected site: a canopy air temperature logger attached to a tree near the canopy rain gauge, a canopy subsurface logger buried underground near the canopy rain gauge, and an open subsurface logger buried near the open rain gauge. However, existing temperature loggers have been reaching the end of their lifespan over the last few reporting periods, resulting in gaps in the types of loggers present at some of the monitored sites (see Table 3.1 for a summary of loggers present at each site for this reporting period). Temperature data are logged hourly and downloaded annually by BEMP staff. BEMP is in the process of phasing out our current model of temperature loggers and installing similar devices with the additional capability of bluetooth connectivity. This shift will make temperature data and visualizations accessible in real time with any authorized bluetooth-capable device, making this dataset interactive on-site for educational purposes, and also making yearly data downloads safer and more efficient.

Complete temperature monitoring methodology can be found online at:

<https://secureservercdn.net/45.40.146.38/659.541.myftpupload.com/wp-content/uploads/2017/09/TempLoggerDownloadandDeploy.pdf>

Table 3.1 List of temperature data loggers present by site.

Site (North to South)	Canopy	Canopy sub	Open sub
Santa Ana	PRESENT	absent	absent
Alameda	PRESENT	absent	absent
RGNC	absent	PRESENT	PRESENT
Route 66	PRESENT	PRESENT	PRESENT
Biopark	absent	PRESENT	PRESENT
AOP	PRESENT	PRESENT	PRESENT
SLO	PRESENT	absent	absent
Los Lunas	PRESENT	absent	absent
Lemitar	absent	absent	PRESENT

Data were run through a visual QA/QC to make sure plots follow the general expected seasonal patterns and historical trends. The data were then checked for the number of NA (missing data points) by site over time and for any points more than three standard deviations (SD) away from the z-score transformed data. The number of data points flagged as outside the three SD were minimal given the volume of data. Data points collected on the same day as the logger was handled for download were eliminated as they did not account for the full day, and as the logger position was disturbed in the download process.

Loggers recording air temperature in the bosque canopy throughout the Middle Rio Grande Valley ranging from the northernmost site at the Pueblo of Santa Ana through the southernmost site at Lemitar recorded a minimum of -16.92°C (Santa Ana on February 4, 2022) and a maximum of 46.25°C (Lemitar on July 21, 2022) for the 2021-2022 reporting period. Four main cold snaps occurred throughout the bosque from December 2021 through March 2022. The warmest air temperatures recorded by canopy loggers for this reporting period's summer season occurred in mid June and late July, with the latter spike reaching the most extreme heat of the year. See figures below for mean daily temperatures captured by canopy, canopy subsurface, and open subsurface loggers for the full 2021-2022 reporting period across all monitored sites.

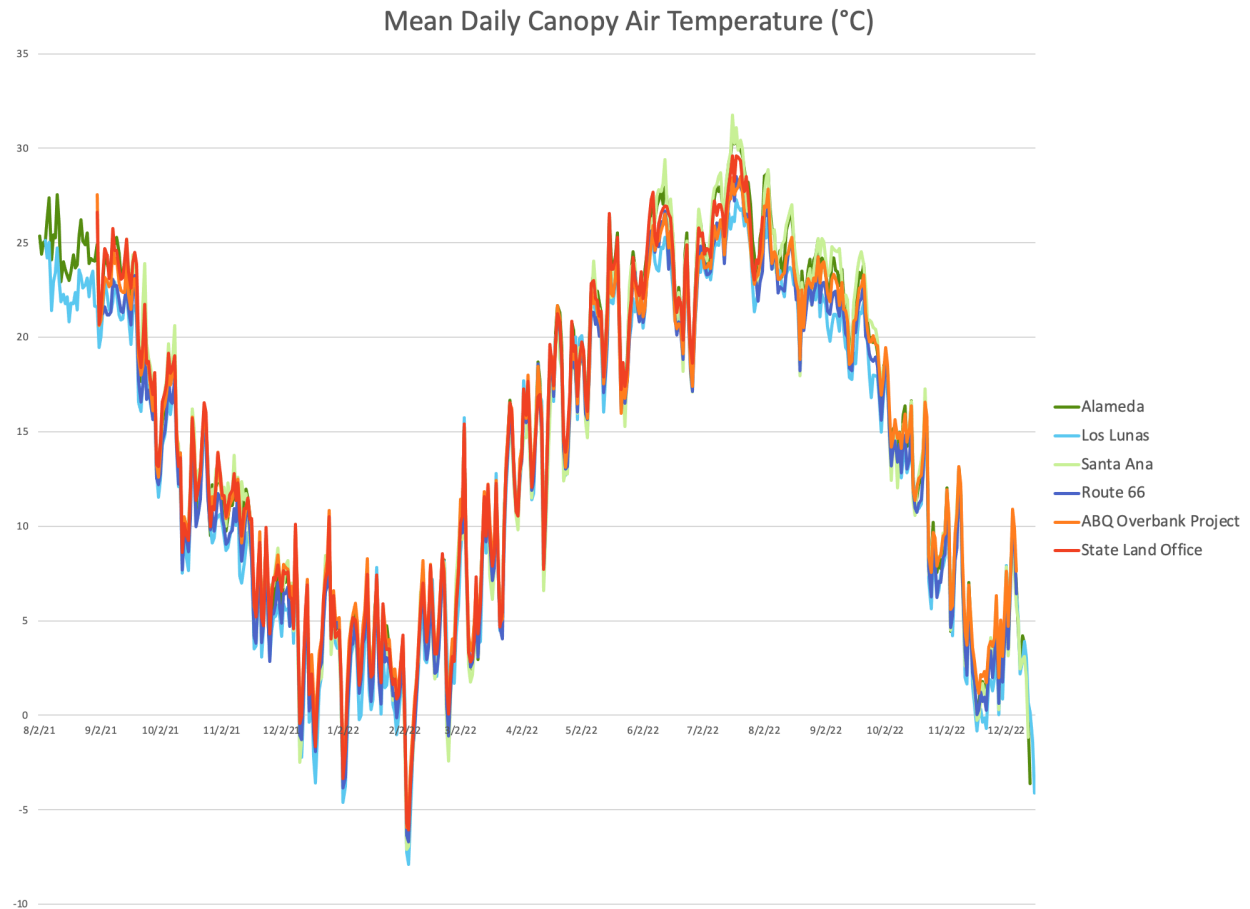


Figure 3.1. Mean daily air temperature (°C) under a canopy from August 2021 through December 2022 at Santa Ana, Albuquerque sites (Alameda, Route 66, Albuquerque Overbank Project, and State Land Office), and Los Lunas.

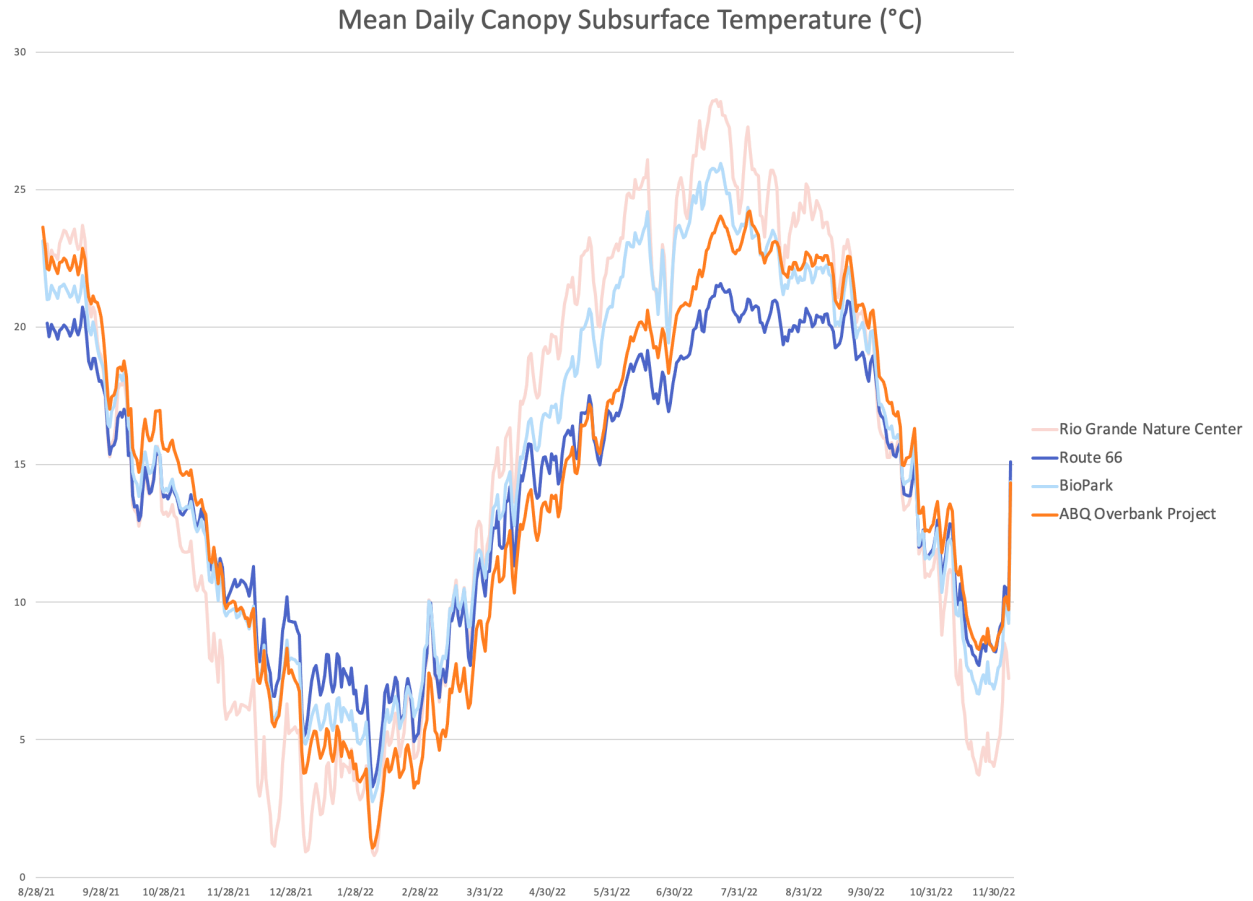


Figure 3.2 Mean daily subsurface temperature (°C) under a canopy from August 2021 through November 2022 at Albuquerque sites (Rio Grande Nature Center, Route 66, Albuquerque Overbank Project, and BioPark).

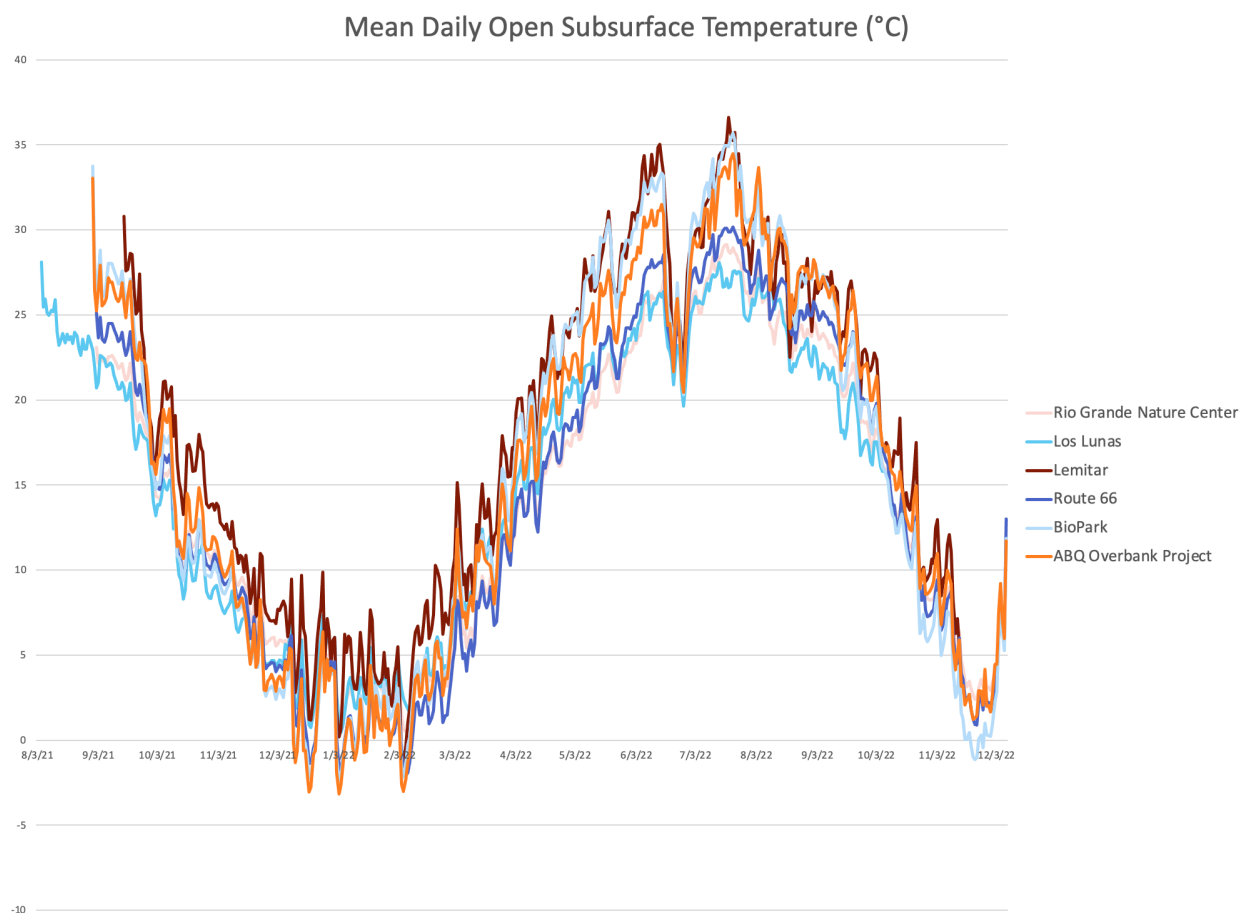


Figure 3.3 Mean daily subsurface temperature (°C) in the open from August 2021 through December 2022 at Albuquerque sites (Rio Grande Nature Center, Route 66, Albuquerque Overbank Project, and BioPark), Los Lunas, and Lemitar.

Sites where all three logger types were present, Route 66 and ABQ Overbank Project, both reveal the same trend of more extreme ground temperature fluctuations in areas clear of tree canopy (Figure 3.4). Open subsurface loggers recorded higher maximums and lower minimums and more closely tracked the air temperature curve captured by the canopy air temperature logger when compared to the subsurface loggers under the tree canopy. This suggests that tree canopy plays a significant insulating role for the ground below.

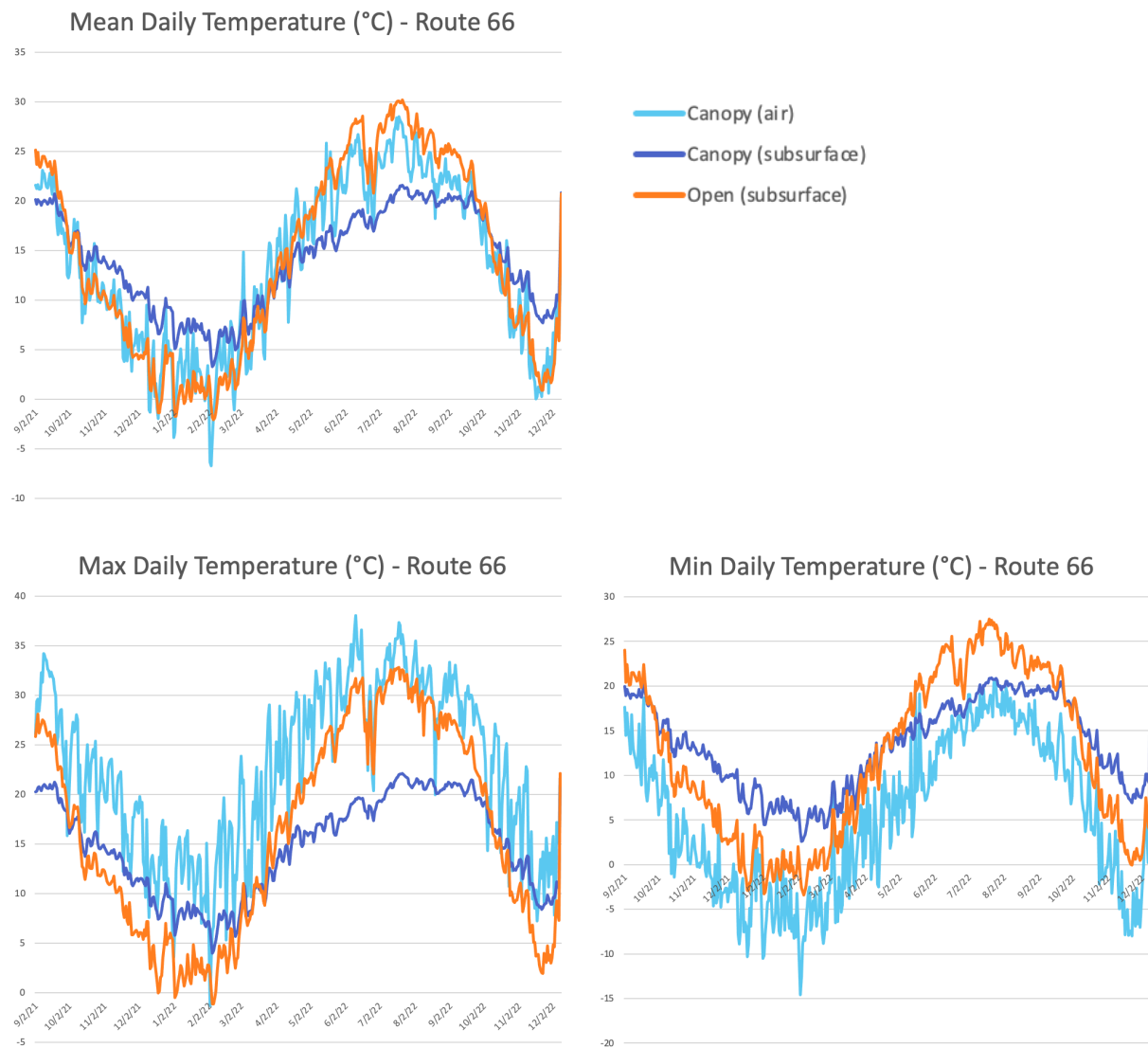


Figure 3.4 Mean, maximum, and minimum daily temperatures recorded across logger locations for Route 66, one of two sites where all three loggers were present for the 2021-2022 reporting period.

4 Precipitation

Precipitation has been measured at all of our sites, but collection of precipitation at Bosque Farms has been halted since August of 2014 due to repeated vandalism (precipitation collection at Valle de Oro was also halted beginning in November of 2017 due to unsanitary conditions, until the site was reestablished in September of 2022). At each site where precipitation is measured, two Tru-Chek precipitation gauges are installed on a post; one under the forest canopy, and one out in the open. Each rain gauge is monitored and emptied by BEMP staff and community scientists once per month. A small amount of oil is added to the empty gauge to prevent evaporation and to ensure capture of the full month's precipitation.

More details on our methods for collecting precipitation data can be found here:

https://github.com/BEMPscience/bemp_data/tree/master/precipitation

Monthly Precipitation for the Year 2022 by Reach

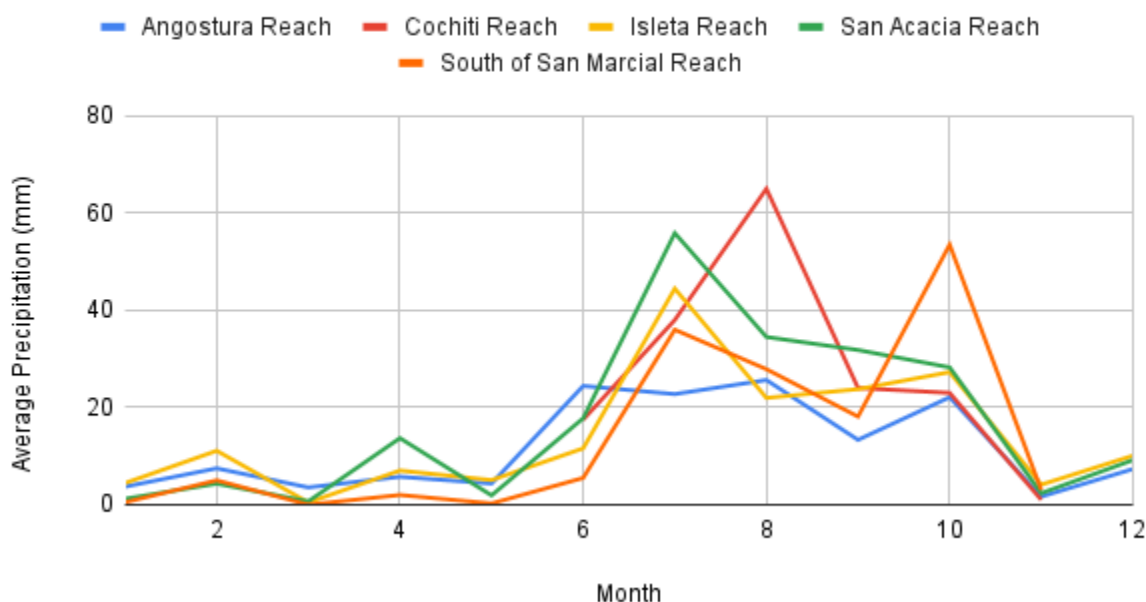


Figure 4.1. Monthly mean precipitation for 2022 by reach.

The highest amounts of precipitation occurred throughout the monsoon season, in July, August, and October (Figure 4.1). This supports the general trend of precipitation shifting towards the later months. In previous years with a spike in precipitation, the peak occurred in July (year 2021), August (years 1999, 2001, 2006, and 2010), September (year 2002), October

(years 2000, and 2017), or has peaked in August and then again in October (years 2005 and 2008).

Comparison Between Precipitation in 2021 and 2022

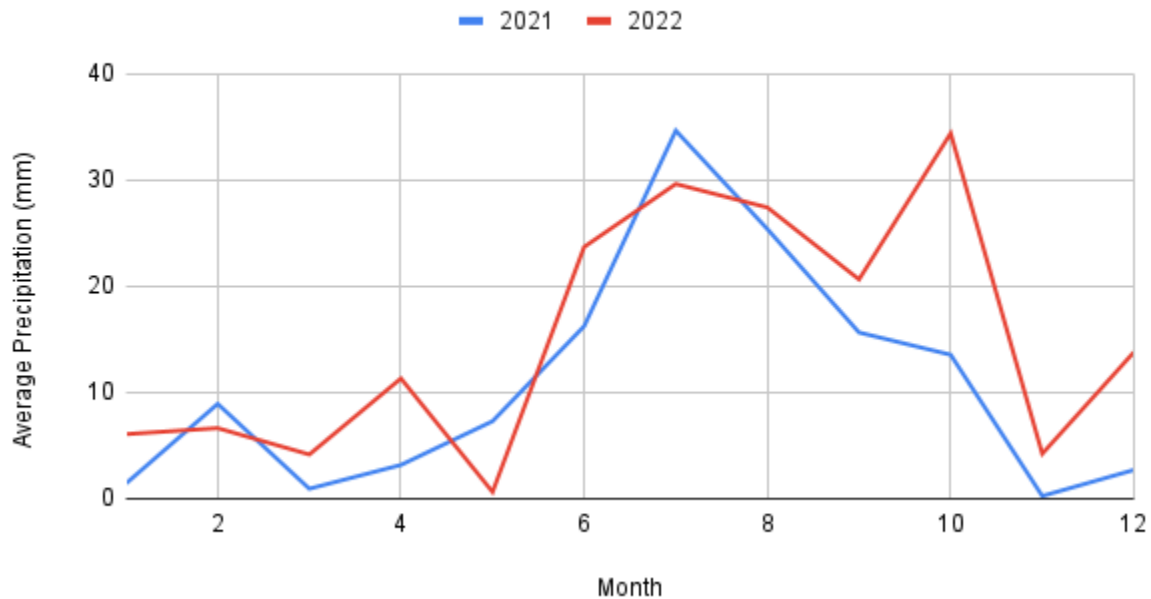


Figure 4.2. Comparison between average precipitation in 2021 and 2022 across all sites.

On average, sites in 2022 had 46.9% more precipitation than in 2021. However, 2022 on average only had 5% more precipitation than the cumulative average of the years 1997 through 2021.

Average Monthly Precipitation in the Angostura Reach

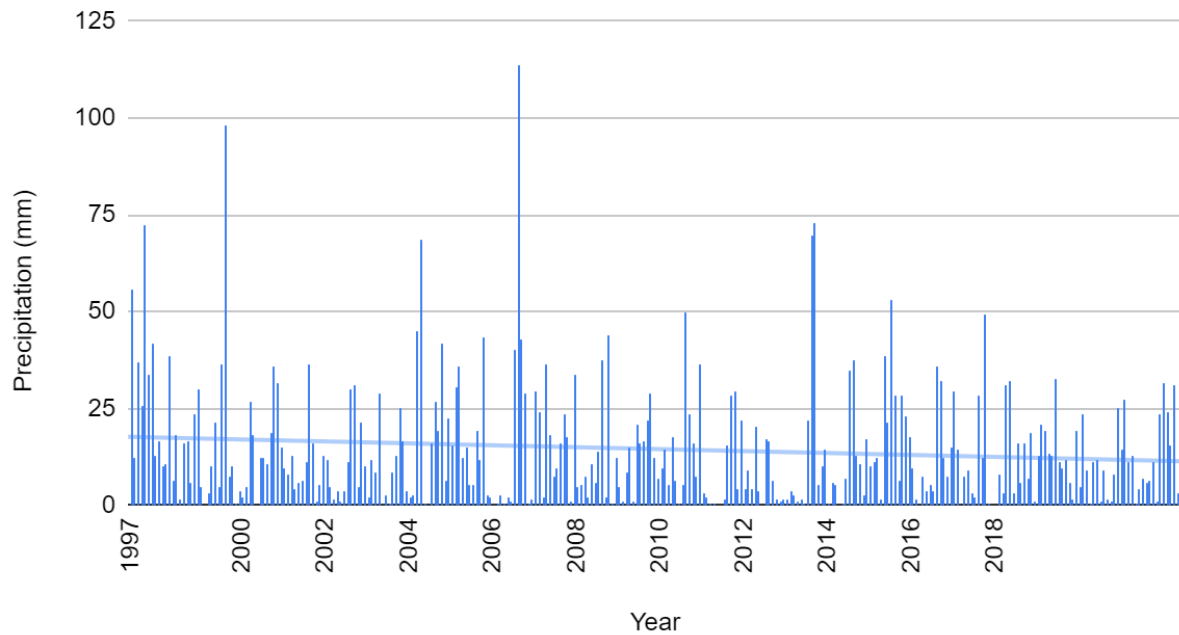


Figure 4.3. The average monthly precipitation for sites south of the Angostura Reach.

Average Monthly Precipitation in the Cochiti Reach

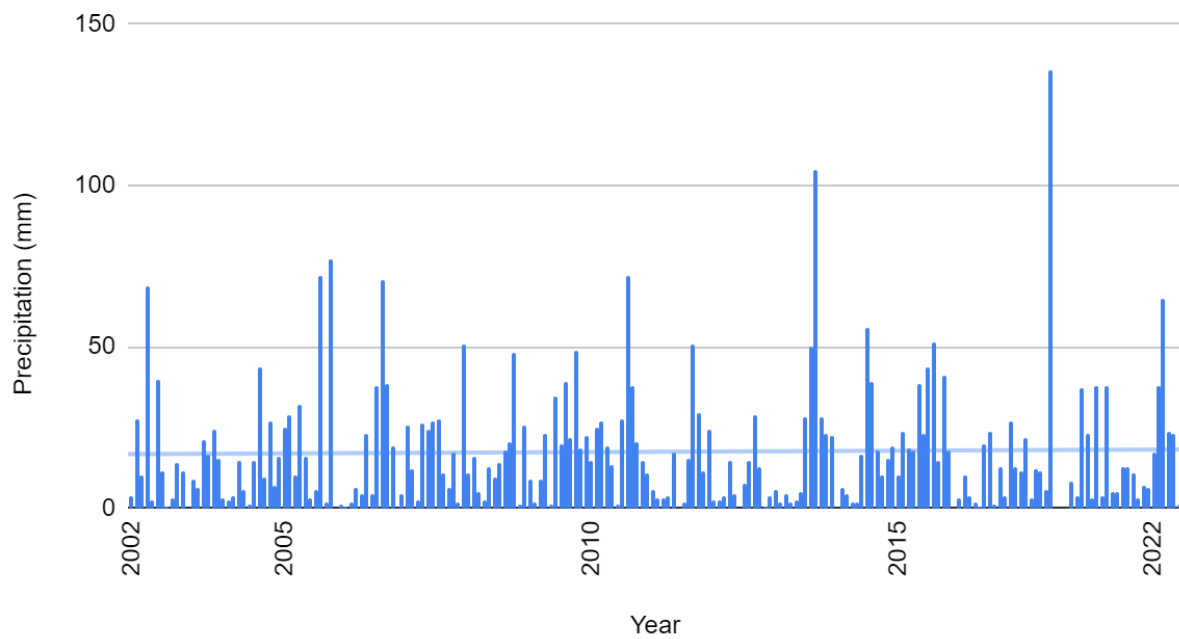


Figure 4.4. The average monthly precipitation for sites in the Cochiti Reach.

Average Monthly Precipitation in the Isleta Reach

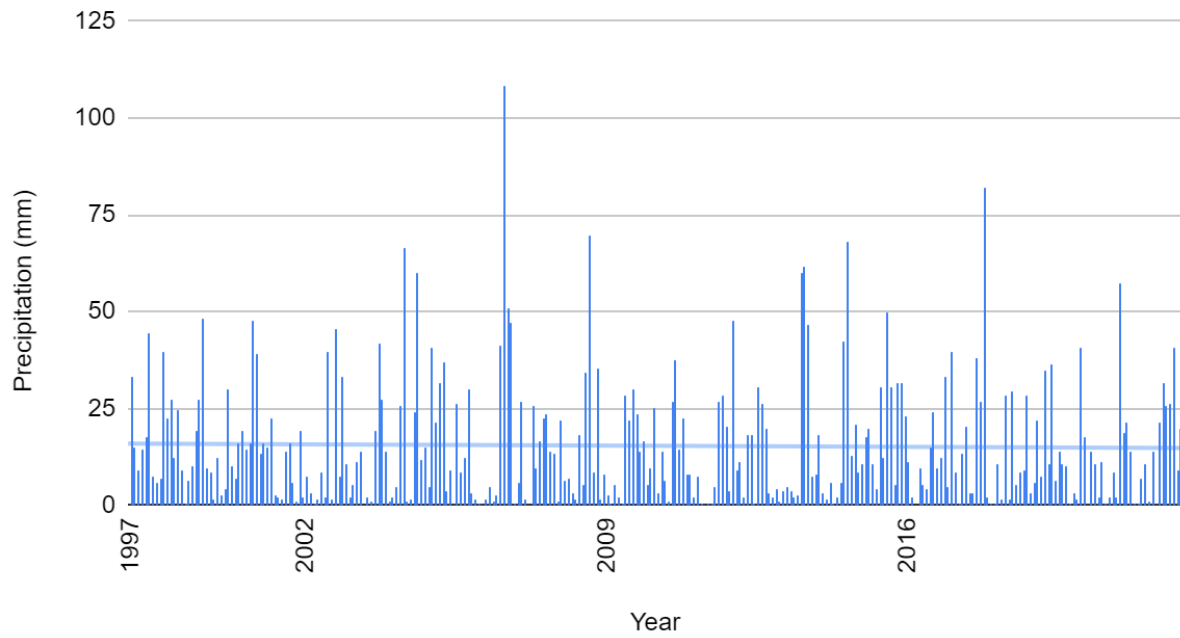


Figure 4.5. The average monthly precipitation for sites south of the Isleta Reach.

Average Monthly Precipitation in the San Acacia Reach

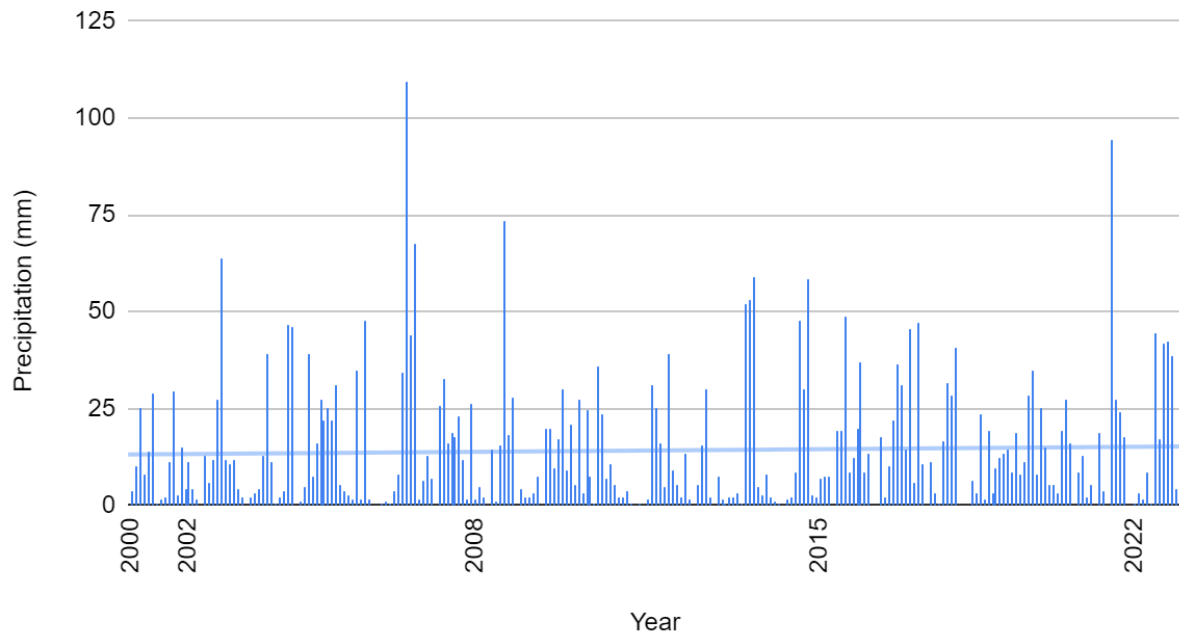


Figure 4.6. The average monthly precipitation for sites in the San Acacia Reach.

Average Monthly Precipitation in the South of San Marcial Reach

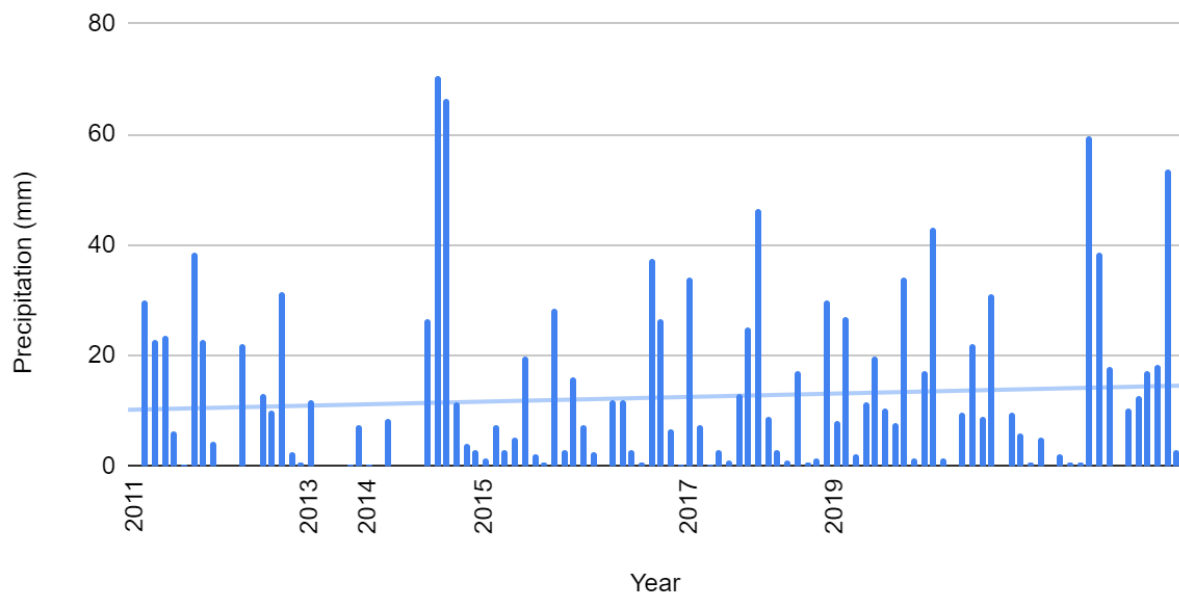


Figure 4.7. The average monthly precipitation for sites in the San Marcial Reach.

For the Isleta and Angostura reaches, there is a slight downward trend in precipitation over time (Figures 4.3, 4.5), while the other reaches either show consistent precipitation (Figure 4.4), or a slight upward trend (Figures 4.6, 4.7). Isleta and Angostura are the reaches that have been monitored the longest and have the most monitoring sites. This makes the trends from those reaches more comprehensive for showing long-term precipitation patterns.

5 Depth to groundwater

Depth to groundwater is monitored at all BEMP sites with the exception of the Pueblos of Santa Ana and Santo Domingo, sites 5 and 24. Groundwater data are collected with permission at the Pueblo of Sandia, but these are proprietary data and requests for groundwater data must go through the Department of Natural Resources at the Pueblo. At all other BEMP sites, five groundwater wells are monitored monthly by BEMP staff along with K-12 students, teachers and UNM interns. Nearby ditches and drains are also monitored during monthly collections. The USGS river flow data are downloaded based on the day of monitoring from the closest gauge to the north of each respective site.

Full monitoring methods can be found at:

https://github.com/BEMPscience/bemp_data/tree/master/depth_to_groundwater_data



Figure 5.1. Spark line plot of the mean monthly depth to groundwater across all BEMP sites. Time from 1997 to 2022 runs across the x-axis and the y-axis is depth to groundwater in cm. Sites are arranged from north to south by center point latitude.

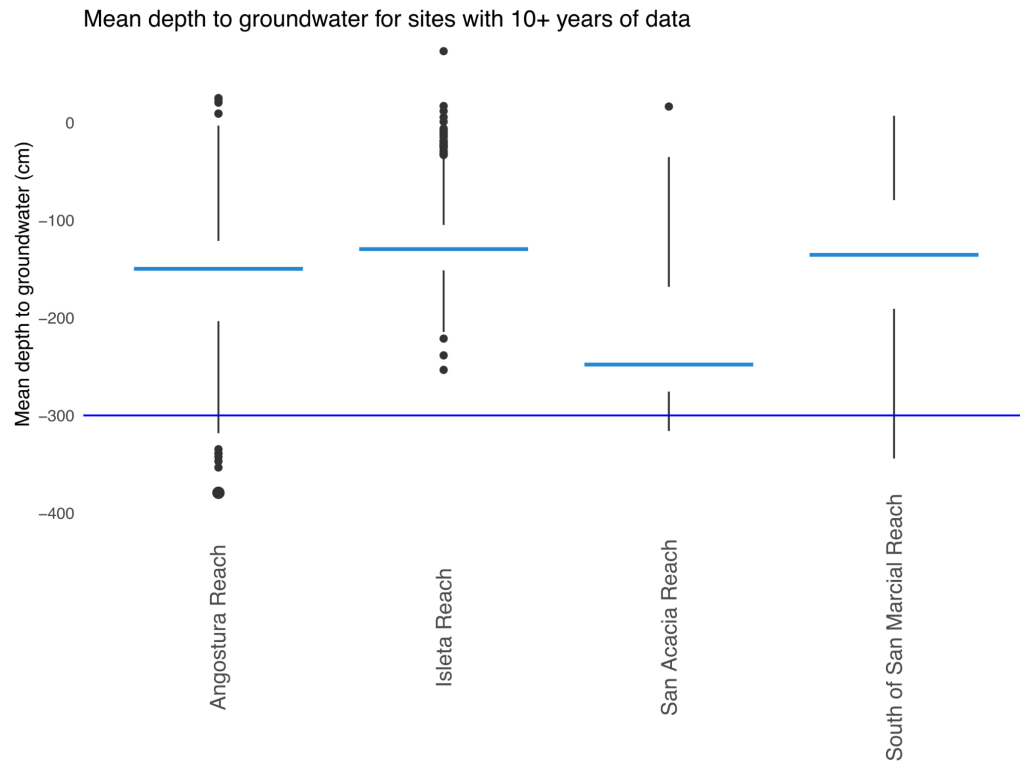


Figure 5.2. Monthly mean depth to groundwater by reach for sites with 10+ years of data. The blue line represents the critical threshold of water depth for mature cottonwood trees.

Cochiti Reach

Depth to groundwater data are unavailable for the Cochiti reach. The Ohkay Owingeh site is inactive. The Santa Ana and Santo Domingo sites are active, but BEMP does not have groundwater wells installed at those locations. Depth to groundwater data are collected at the Sandia site, but are proprietary. Requests for these data must go through the Department of Natural Resources at Sandia Pueblo.

Angostura Reach

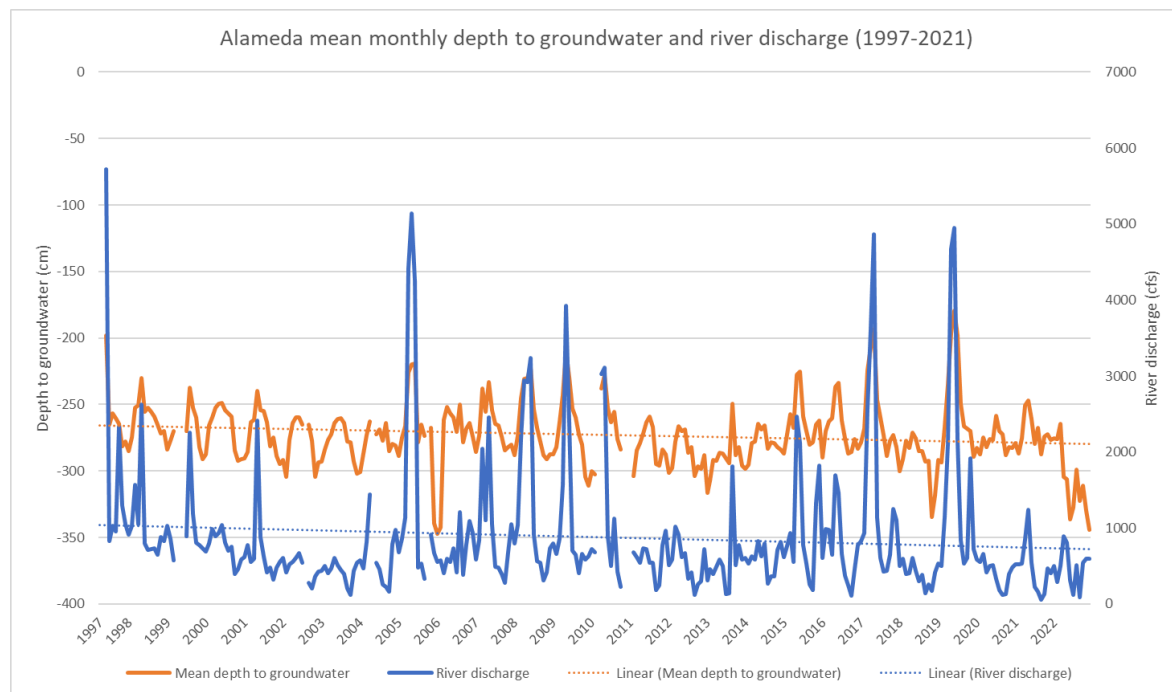


Figure 5.3. Monthly mean depth to groundwater and river discharge (USGS Central gauge) for the Alameda site (1) from 1997-2022.

Well data were not collected for the Calabacillas (11) site from January- April 2022, and HCC (8) site from January-July 2022 due to a funding gap. Valle de Oro (28) was reestablished in September 2022 after a three-year gap in collections. Well data are consequently difficult to meaningfully analyze.

Mean depth to groundwater within the Angostura reach was shallower at six sites, the same at seven sites, and deeper at three sites when compared to 2021 well data. For the sites that were shallower, none exceeded 15cm shallower than the prior year. For the sites that were deeper, two of the three were less than 10 cm deeper on average than in 2021. Notably, however, the Alameda (1) site was a full 30 cm deeper on average. Although 2022's mean river discharge was higher than the previous year, the diversion dam just north of the Alameda site may have affected river flow to that area.

On a broader time scale, depth to groundwater within the Angostura reach has remained consistent. Some sites show slight downward trends in depth to groundwater, but this is closely correlated to the similarly downward trending river discharge rates. Despite the relatively consistent mean depth to groundwater across all sites in the Angostura reach, the minimum, as

well as several outlier data points, lay below the mature cottonwood root threshold (Figure 5.7).

Isleta Reach

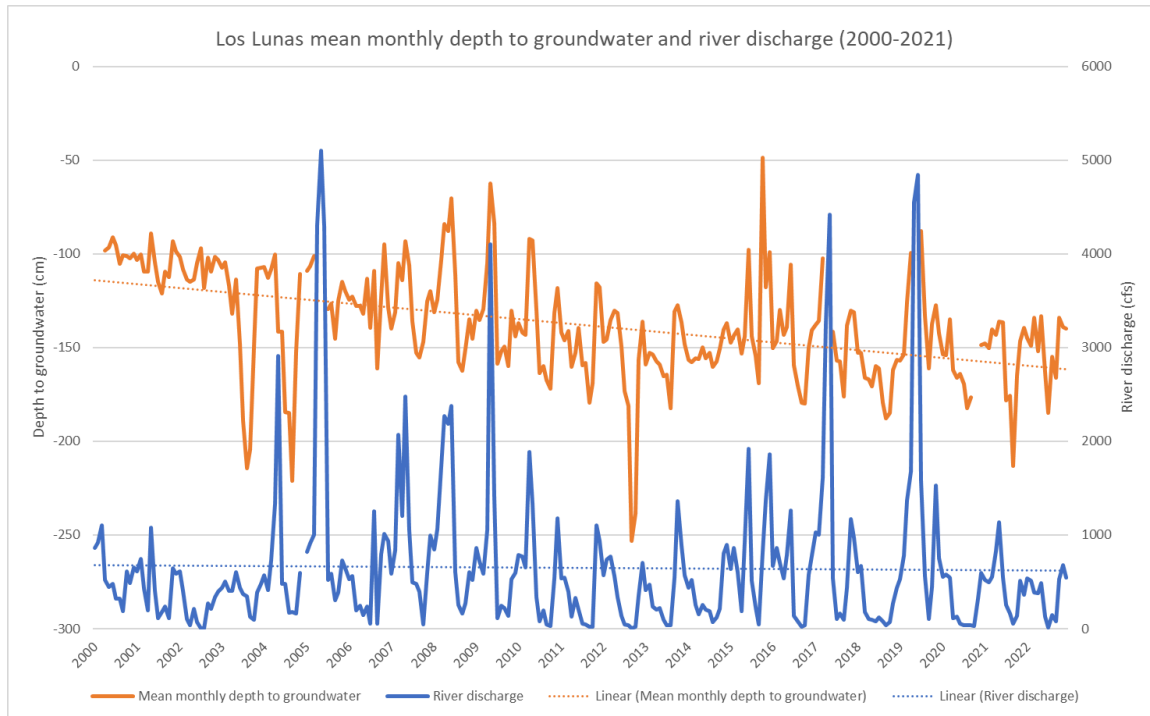


Figure 5.4. Monthly mean depth to groundwater and river discharge (USGS Bosque Farms gauge) for the Los Lunas site (3) from 2000-2022.

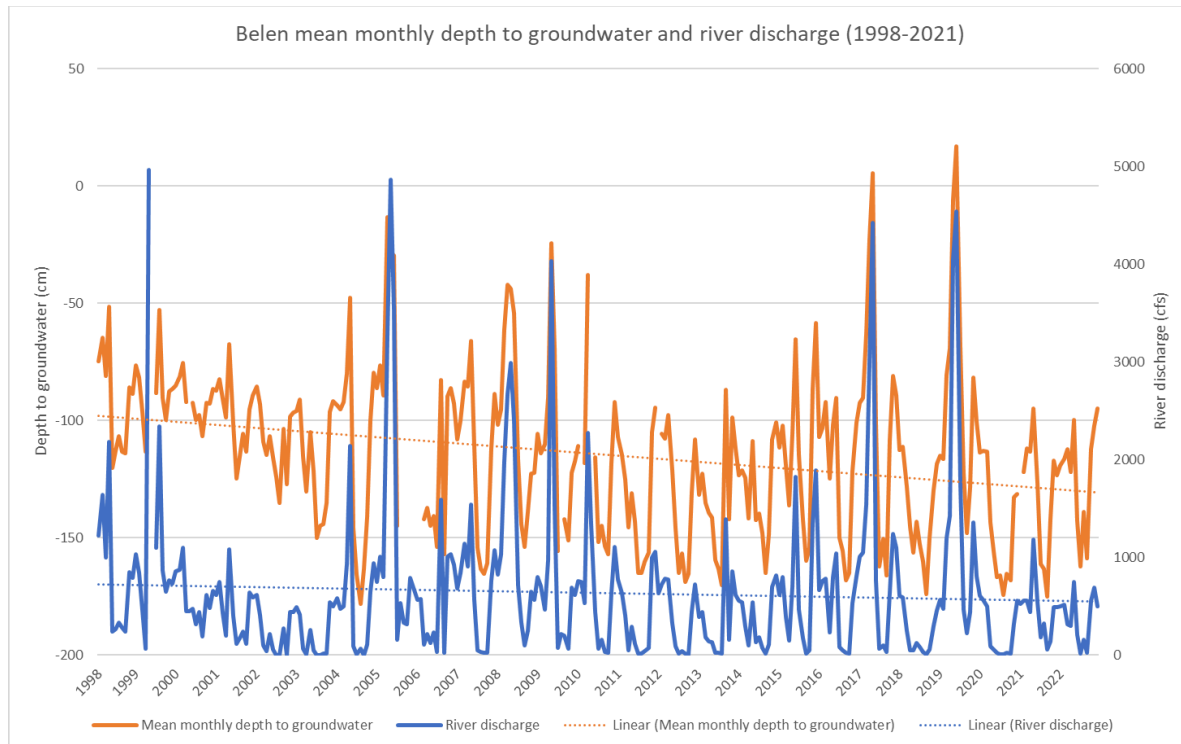


Figure 5.5. Monthly mean depth to groundwater and river discharge (USGS Bosque Farms gauge) for the Belen site (4) from 1998-2022.

Well data for the Reynold's Forest site (18) were not collected from January-April 2022 due to a funding gap.

Compared to 2021 depth to groundwater, depth to groundwater in sites within the Isleta reach was 6-8 cm higher in 2022. As with the Angostura reach, this is tightly correlated to slightly increased river flows compared to the previous year.

Although depth to groundwater measurements within the Isleta reach are consistently shallower than the other reaches, five of the eight sites are trending downwards. Analysis of the two oldest sites (Los Lunas and Belen) highlight this downward trend as both sites show a roughly 25-50cm drop in mean monthly depth to groundwater over time (Figures 5.4 and 5.5). Note that if years with abnormally high river discharge are removed from the data, river flow also trends noticeably downward.

San Acacia and South of San Marcial Reaches

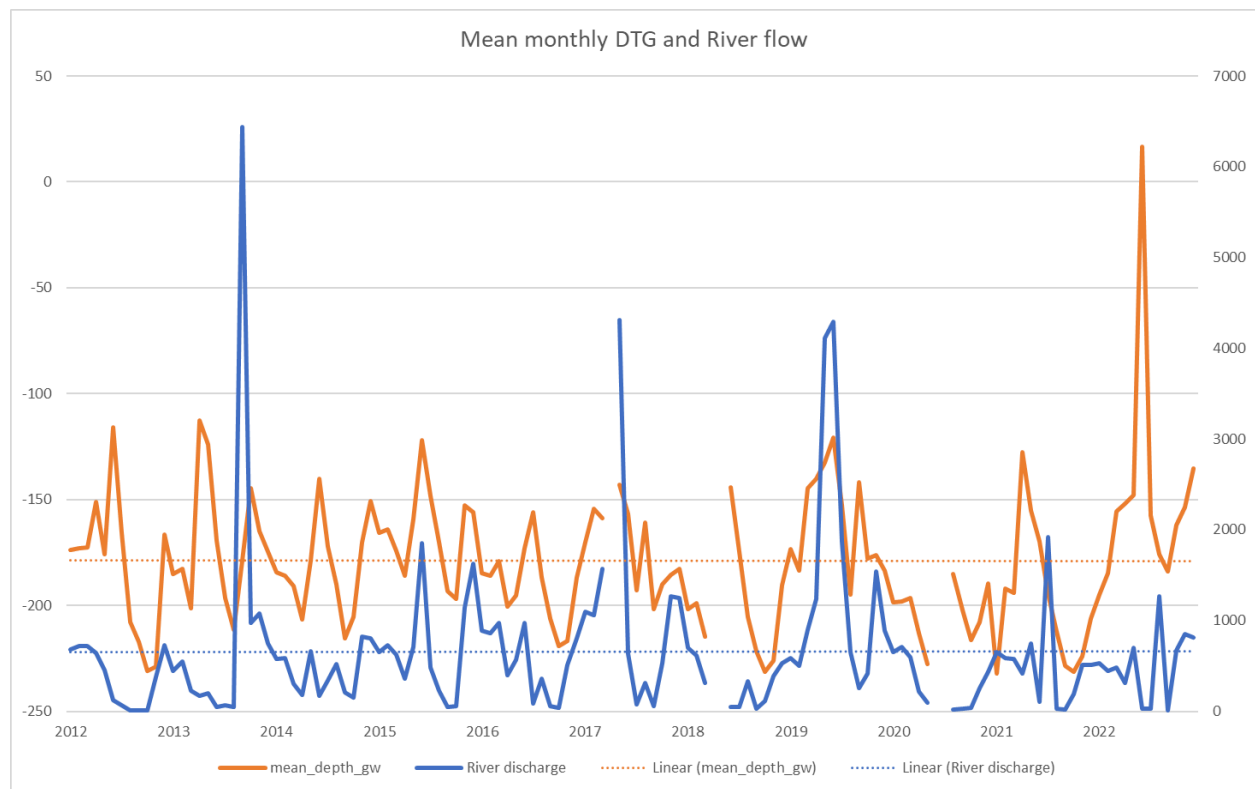


Figure 5.6. Monthly mean depth to groundwater and river discharge (USGS San Acacia gauge) for the Sevilleta site (14) from 2012-2022. Note the flooding event in June 2022 which does not correlate with river flow.

Well data for the Mesilla Valley Bosque State Park (26) were not collected from October 2021-April 2022 due to a funding gap.

In 2022, there was an 11 cm drop in depth to groundwater at Lemitar (7). Although this site is outside of the levee system, and thus is not as responsive to changes in river flow, the river flows in that area were on average lower than in 2021. Sevilleta experienced a flooding event in June 2022, likely due to rain events pooling at the site. Although BEMP did not catch the peak water level, when the site was collected water was about 16 cm above ground level (Figure 5.6).

Sevilleta and Lemitar mean monthly depth to groundwater values have remained consistent over the last decade. Bosque del Apache seems to be trending downwards, but inconsistent measurements due to flooding, well drying, and the relative newness of the site may skew the

mean data for the reach. Mesilla Valley Bosque State Park groundwater levels have remained consistent as well.

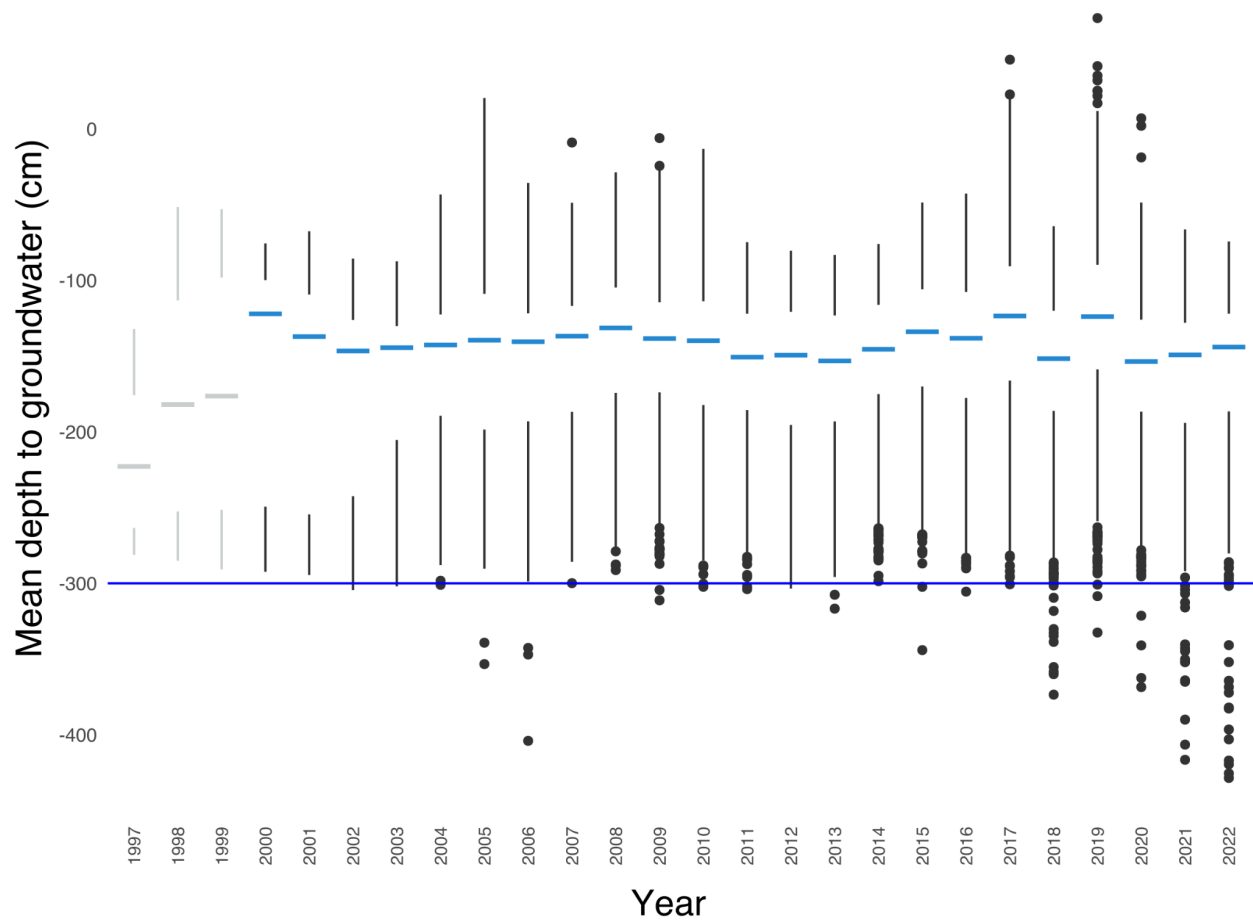


Figure 5.7. Boxplot of the mean monthly depth to groundwater from 1997 to 2022. The blue line is the critical threshold of ground water depth for mature cottonwood trees. The means are the pale blue lines and the black dots are valid data but outliers. Both changes in the mean and variance of the shallow riparian aquifer are impacting our local bosque.

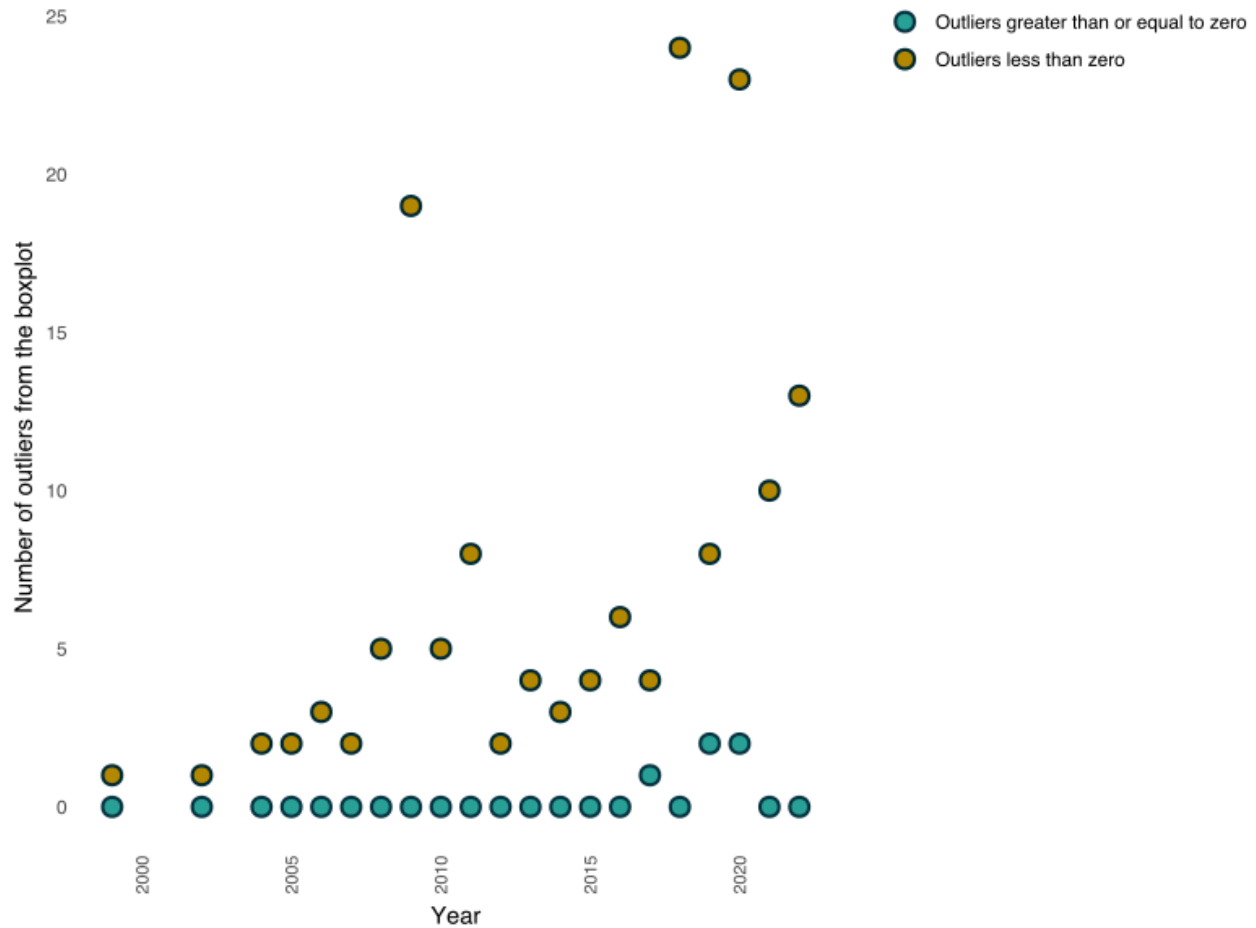


Figure 5.8. Boxplot outliers from the monthly mean depth to groundwater. The outliers (1.5 times the interquartile range) less than zero are more negative depths to groundwater, or drops in the water table. Positive outliers represent response to flood events. The increase in outliers from year to year is unstable which makes building models to predict riparian response to changing water levels more difficult.

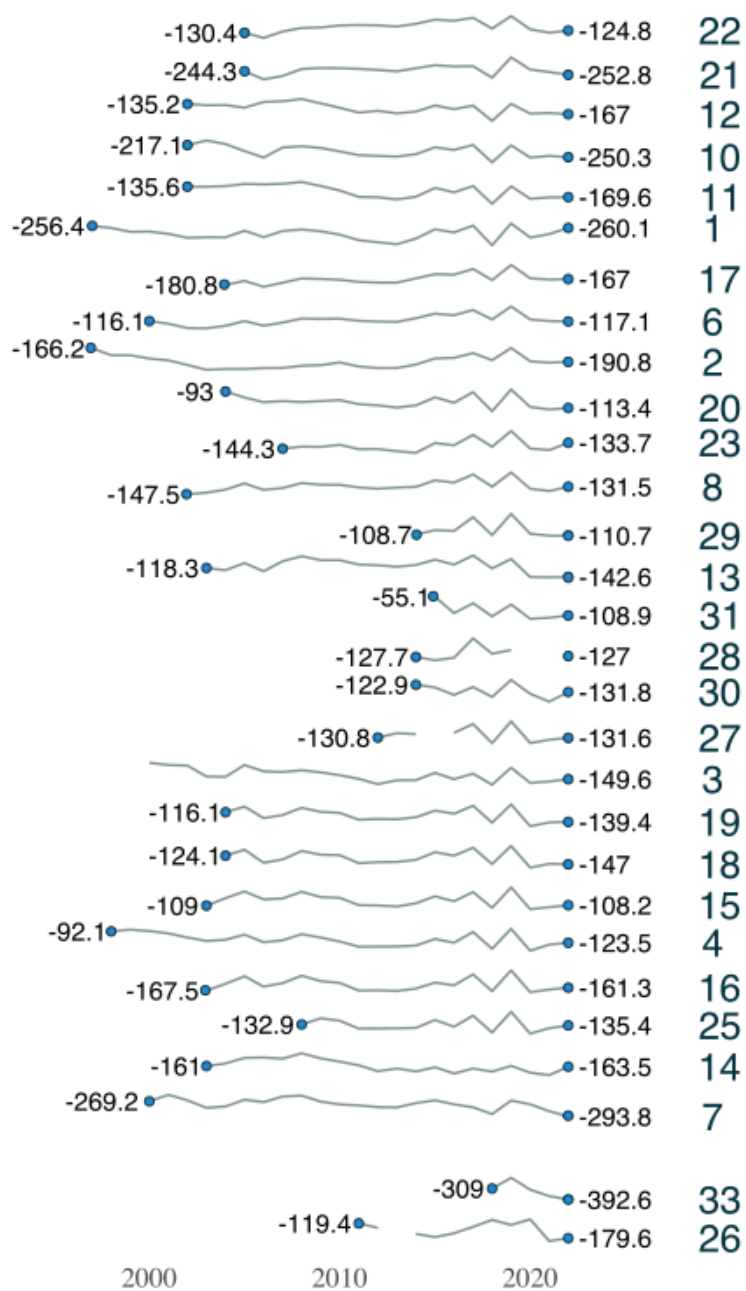


Figure 5.9. Sparklines of mean annual depth to groundwater from 1997 to 2022. Missing values are represented by either a missing circle or a break in the line. Sites are arranged from north to south by site number (starting in Albuquerque).

Changes in mean and variance of depth groundwater in the shallow riparian aquifer.

BEMP is the education outreach Schoolyard Program of the Sevilleta Long Term Ecological Research (LTER) site (National Science Foundation DEB 1655499, LTER: Sevilleta (SEV) Site: Climate Variability at Dryland Ecotones, Lead PI Rudgers). As such, BEMP has adopted the mean-variance approach (<https://sevlter.unm.edu/mean-variance-experiment/>) to looking at BEMP data. The rapid changes in the mean and variance brought on by human-caused climate change may drive ecosystems into transition (riparian to semi-arid or arid). Figure 5.10 shows the mean annual depth to groundwater at Alameda in the Angostura Reach with its percent change from year to year. Angostura Reach is a losing reach with much deeper depth to groundwater in the shallow riparian aquifer than many of our sites in the Isleta Reach. Note that in Alameda (Figure 5.10), the change of percent change is between -15 to +15 which is relatively stable compared to Belen (Figure 5.11).

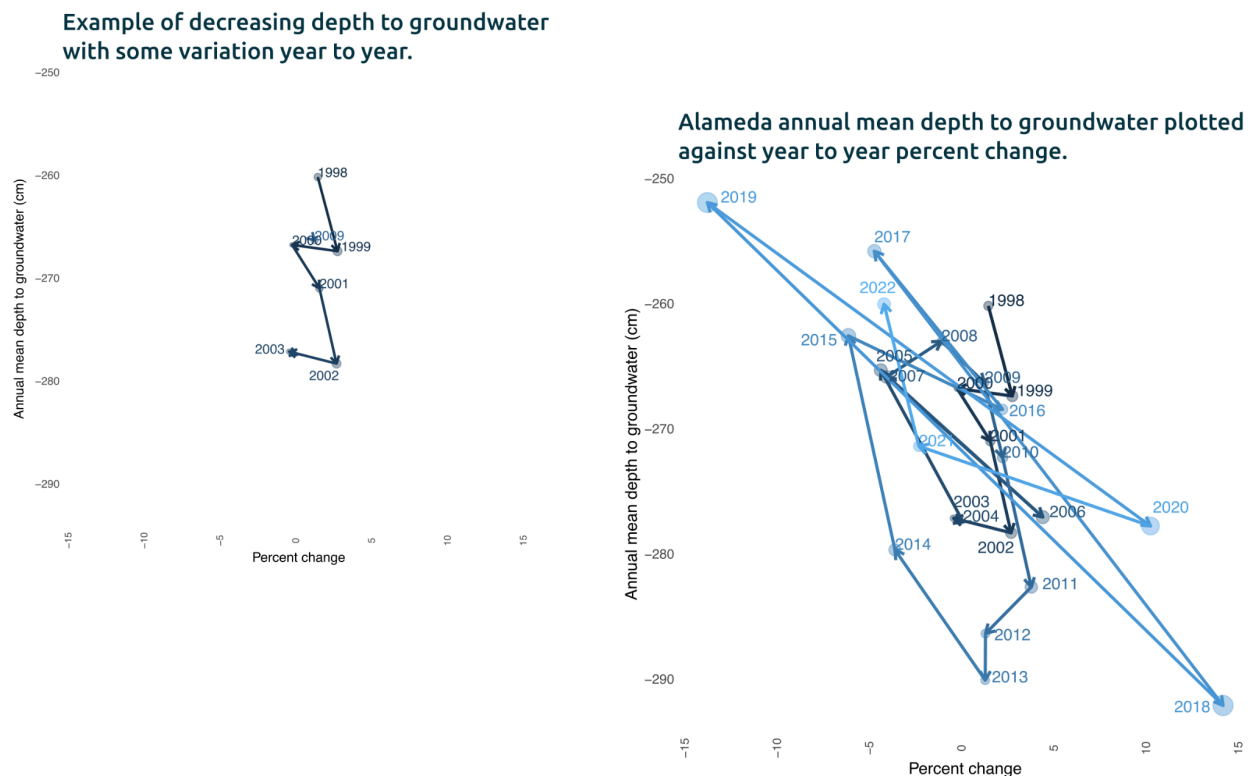


Figure 5.10. The mean depth to groundwater at Alameda (Angostura Reach) plotted against the percent change of depth from year to year (first derivative, f'). Note that after 2003 the year to year variation becomes increasingly more chaotic.

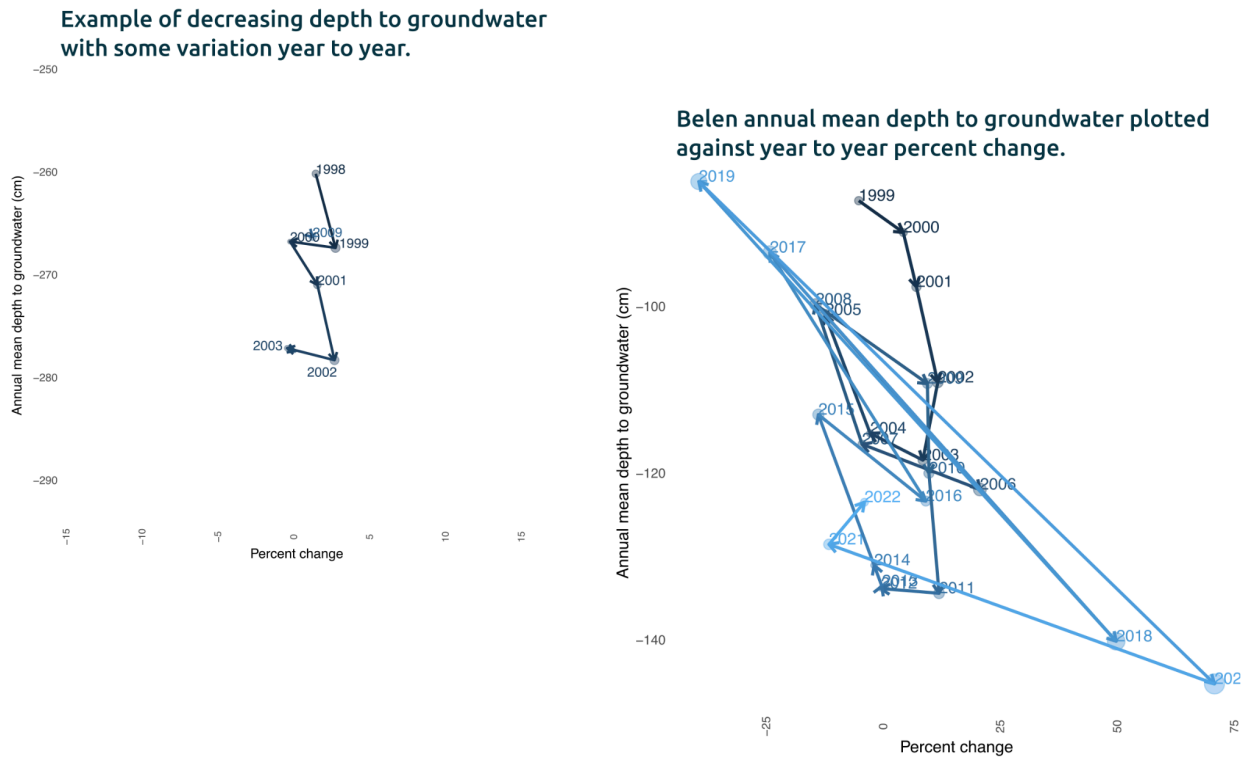


Figure 5.11. The mean depth to groundwater at Belen (Isleta Reach) plotted against the percent change of depth from year to year (first derivative, f'). Note that after 2017 the year to year variation becomes increasingly more chaotic.

Figure 5.11 is in the Isleta Reach at the Belen BEMP site. Many BEMP sites in the Isleta reach shift from gaining reach to losing reach, depending on river flow. The depth to groundwater is much closer to the surface; however, the variance at the Belen sites is radically different. The variance ranges from -30 to $\sim +75$, although the increase in variance didn't occur until 2017 (Figure 5.11).

Understanding the chaotic changes in mean and variance of the depth to the shallow riparian aquifer is critical to gain insight into how our local ecosystem and ecotones will respond to increasing pressures from climate change.

6 Water Quality

BEMP is funded by The Middle Rio Grande Stormwater Quality Team to monitor the Albuquerque (Angostura) Reach of the Rio Grande for the bacterial pathogen *Escherichia coli* and other field parameters. Parameters measured include dissolved oxygen (DO), specific conductance, pH, turbidity, and water temperature. Although the EPA and State of New Mexico recognize the upper limit for *E. coli* in primary contact water as 410 MPN/100 mL, limits set by the Pueblo of Isleta, just south of Albuquerque, are 88 MPN/100mL due to special usage. For this reason it is essential to understand how these parameters change as the river flows through and out of Albuquerque throughout the year. Sampling occurred quarterly in 2022 (January, April, July, and October). Sampling methodologies, sample sites (Figure 6.1), and results are further detailed in the 2022 Annual Stormwater Quality Technical Report, available on request.

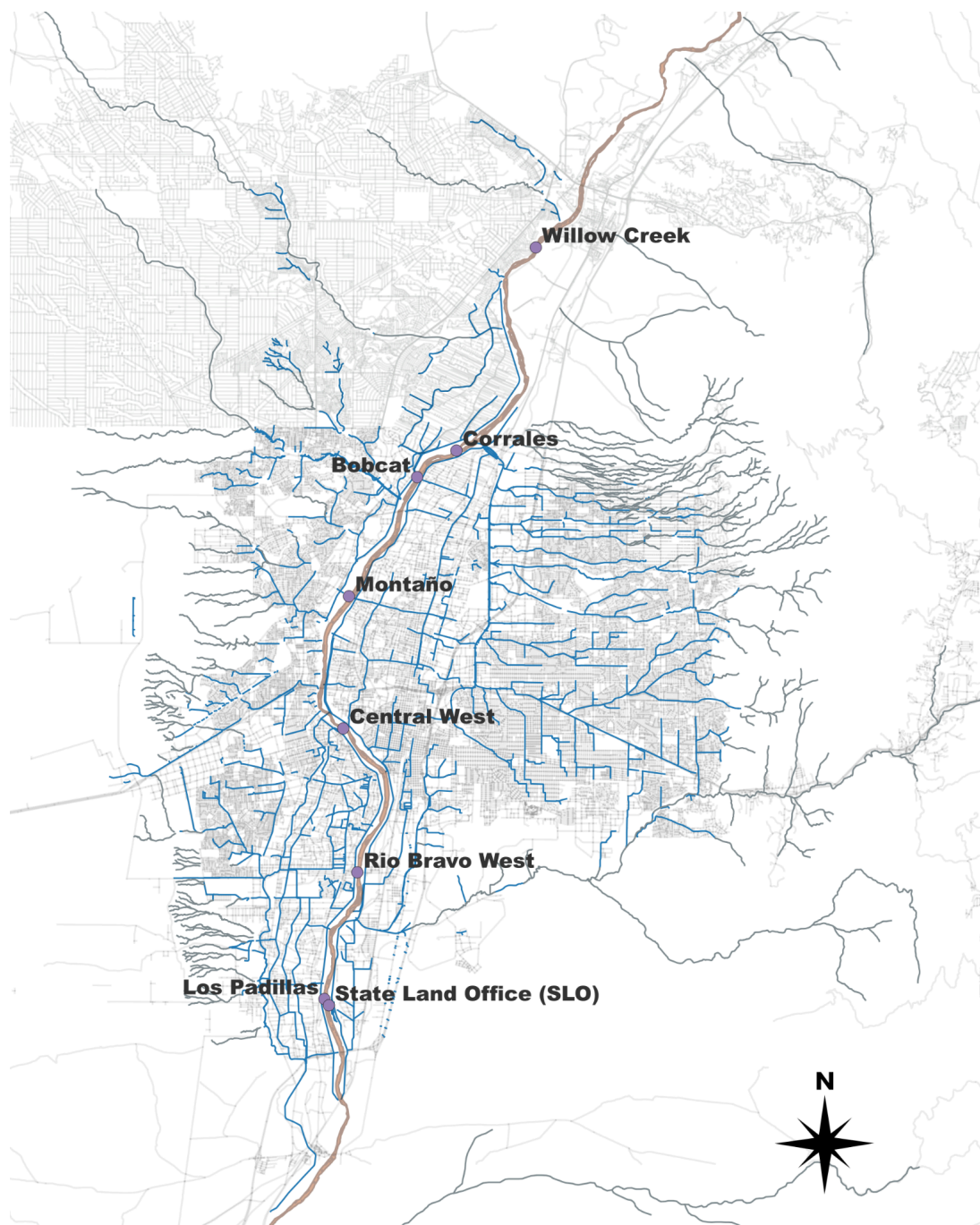


Figure 6.1. Sampling locations for 2022. Additional GIS layers include arroyos, drains and ditches, city streets, and river center.

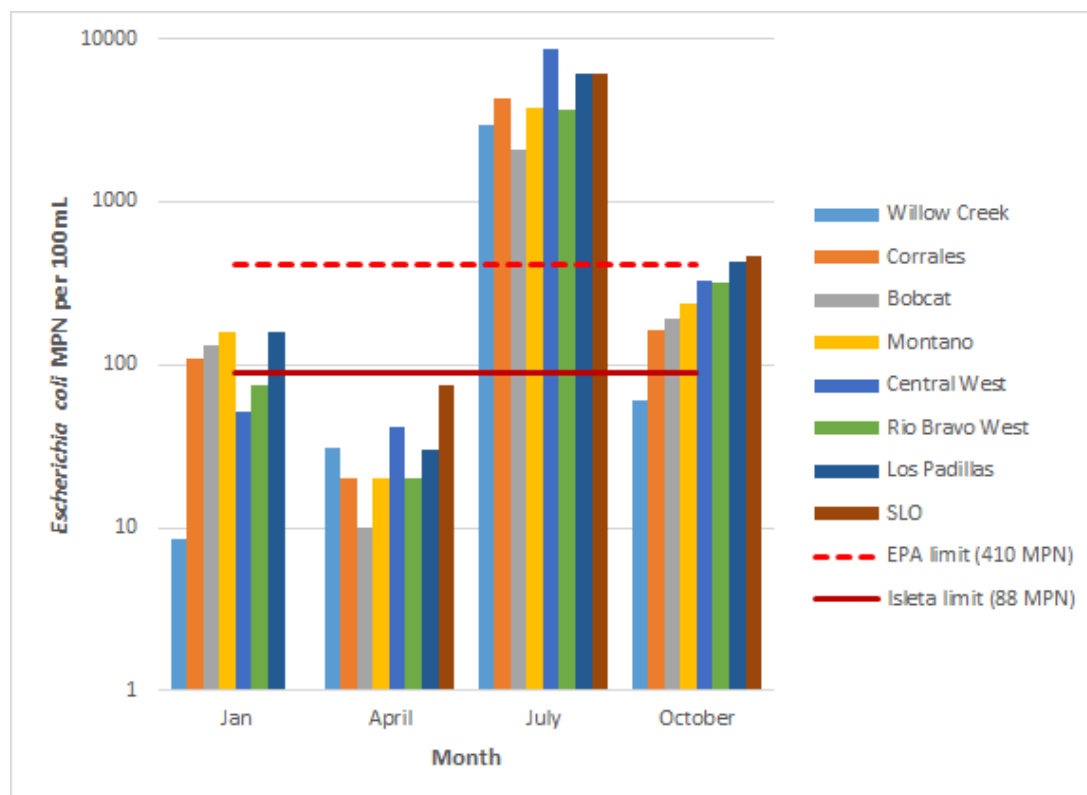


Figure 6.2. *Escherichia coli* MPN/100mL on Log 10 scale at sampling sites across months.

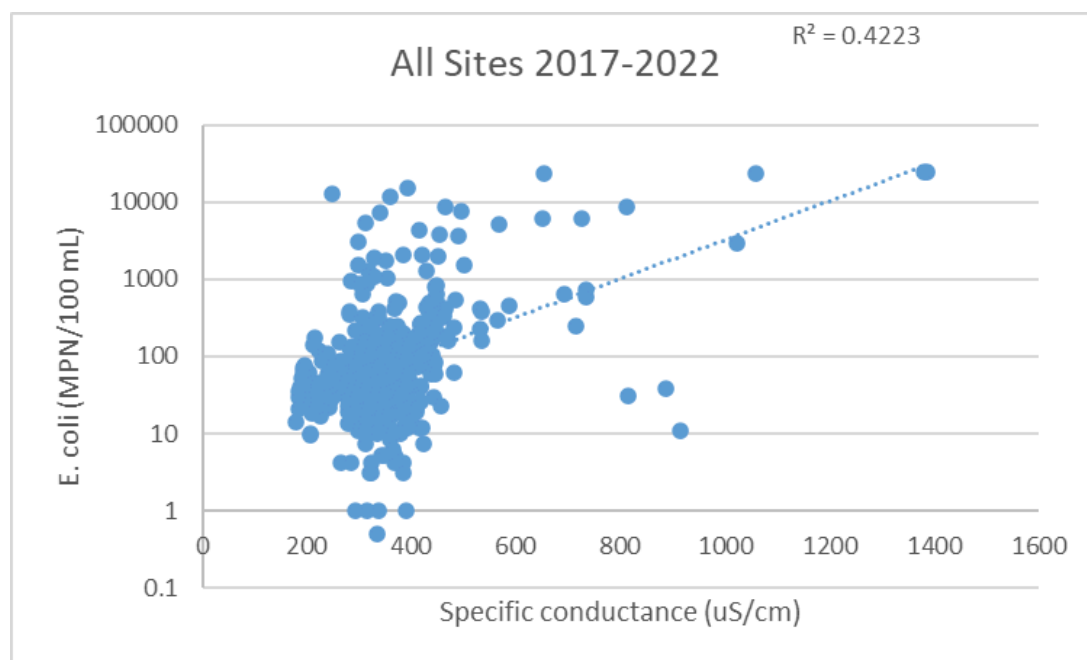


Figure 6.3. Log scale of *E. coli* vs. specific conductance for all sites from 2017-2022.

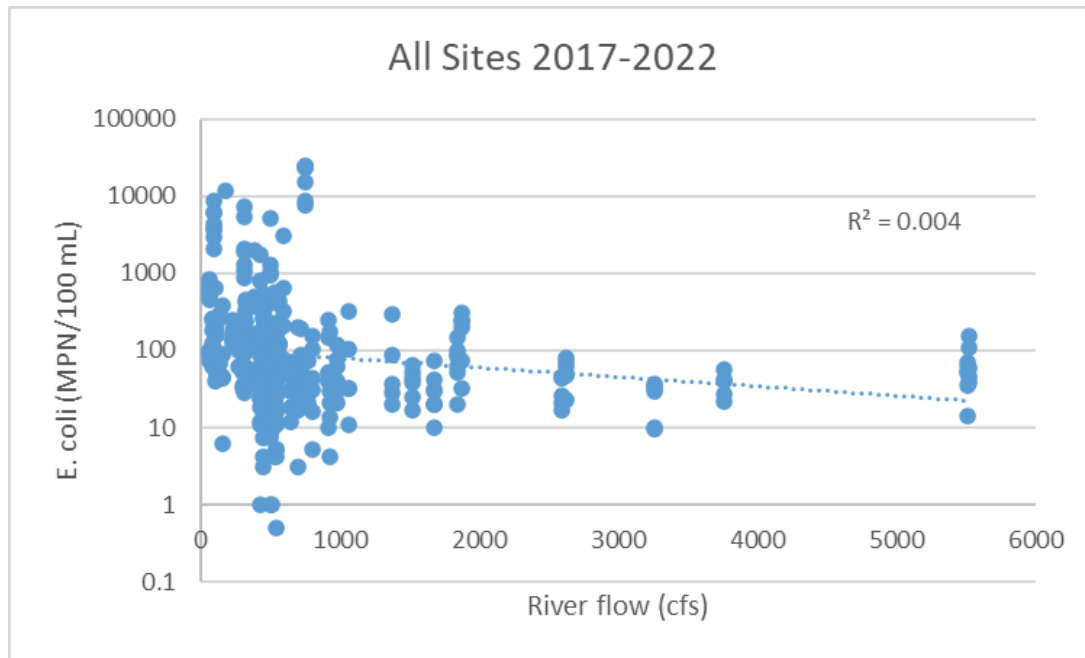


Figure 6.4. Log scale of *E. coli* vs. river flow for all sites from 2017-2022.

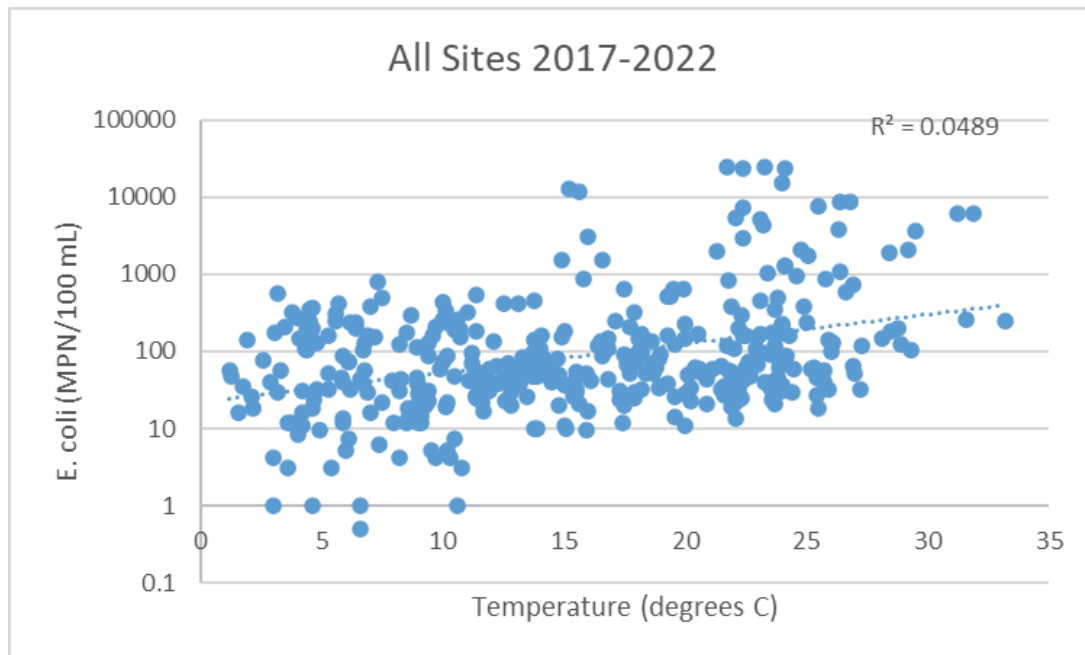


Figure 6.5. Log scale of *E. coli* vs. water temperature for all sites from 2017-2022.

Escherichia coli levels exceeded the desired limit of 88 MPN/100 mL on three of the four sampling dates (January, July and October) (Figure 6.2). *E. coli* exceeded the EPA and state limits of 410 MPN/100 mL on two of those sampling dates in July and October (Figure 6.2). In July, samples exceeded both desired limits (88 MPN/100mL) and EPA limits (410 MPN/100mL) at all sampling locations. The highest *E. coli* level recorded for July was at Central West with 8,664 MPN/100mL. The extremely high *E. coli* levels seen in July correspond with thunderstorms that occurred in the days preceding the collection. These rain events were responsible for the first rewetting of the Rio Grande after days of river drying in the Albuquerque stretch of the Rio Grande, demonstrating how precipitation and runoff events can dramatically impact *E. coli* in the Rio Grande. Exceedances disproportionately occurred in southern sites compared to northern sites, with SLO and/or Los Padillas exceeding the *E. coli* limits in all months that exceedances occurred (Figure 6.2).

Specific conductance has the strongest correlation with *E. coli* levels (R^2 0.4223, Figure 6.3). This relationship is also seen at the site level, though can be driven by outliers. *E. coli* data from this year had a correlation with river flow with a $0.407 R^2$, but when analyzing trends across all sites from 2017-2022, the R^2 was 0.004 (Figure 6.4). Temperature is also a poor predictor of *E. coli* when used alone (Figure 6.5).

7 Litterfall and Vegetation Cover

Litterfall is any plant material that falls to the ground. BEMP litterfall data are categorized into leaves, reproductive parts, and wood from dominant tree species. It is collected monthly and then dried for 48 hours before being sorted and weighed. Litterfall is used to gauge plant productivity (leaves), reproductive effort (buds, flowers, seeds), and stress or senescence (wood).

Full monitoring methods can be found at:

https://github.com/BEMPscience/bemp_data/tree/master/leaf_litterfall

Vegetation cover surveys are conducted in August-September each year by a team of botanists and BEMP staff. Line intercept methods are used to monitor plant species along ten 30 meter transects at 27 sites. Herbarium work (identification of species) has been completed for 2022 data, which are now being entered and QA/QCed. Data through 2021 are included in this report.

Full monitoring methods can be found at:

<https://secureservercdn.net/45.40.146.38/659.541.myftpupload.com/wp-content/uploads/2016/01/vegetation-monitoring-directions.pdf>

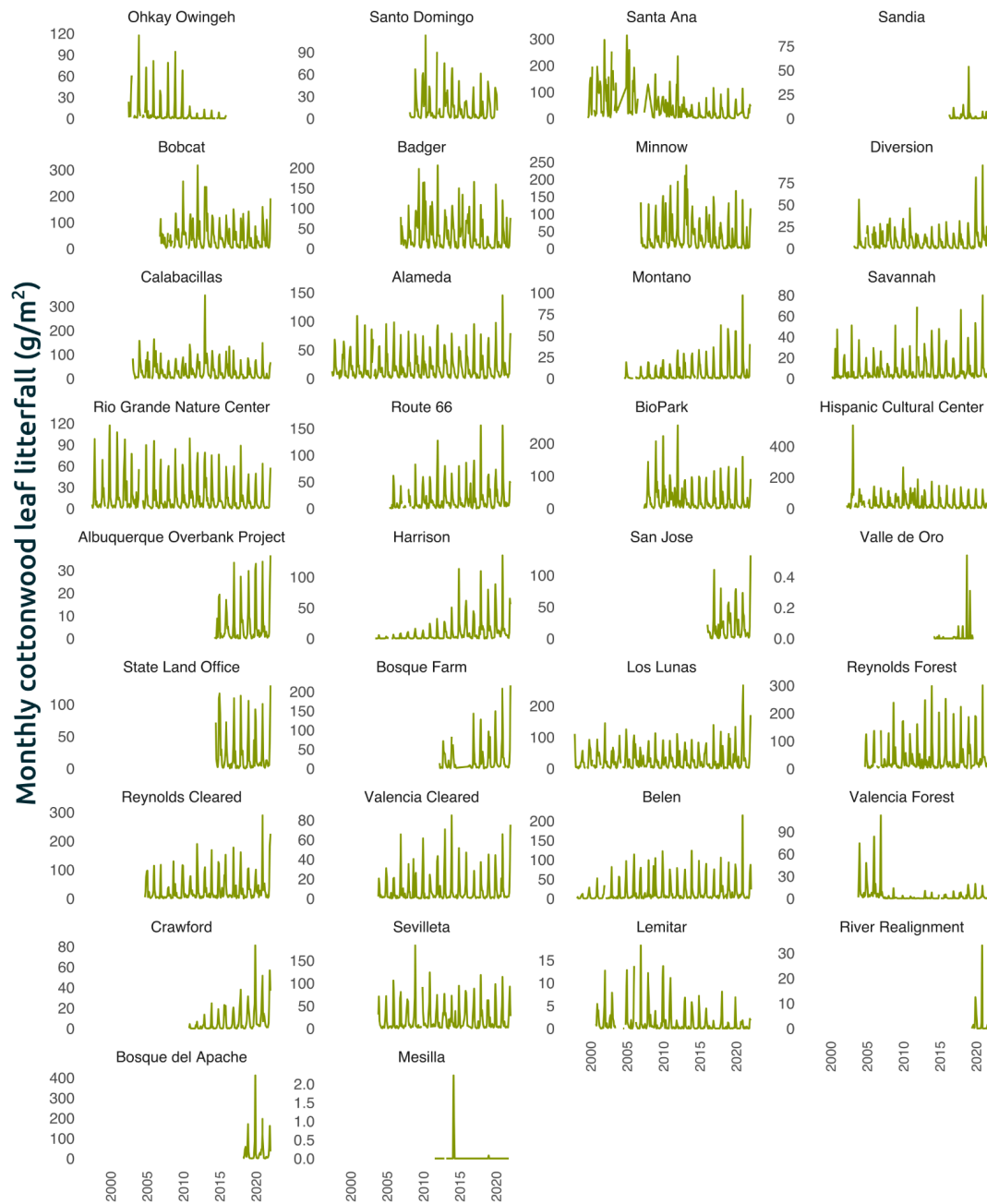


Figure 7.1 Monthly cottonwood litterfall (g/m^2) shown across years at each site (listed north to south).

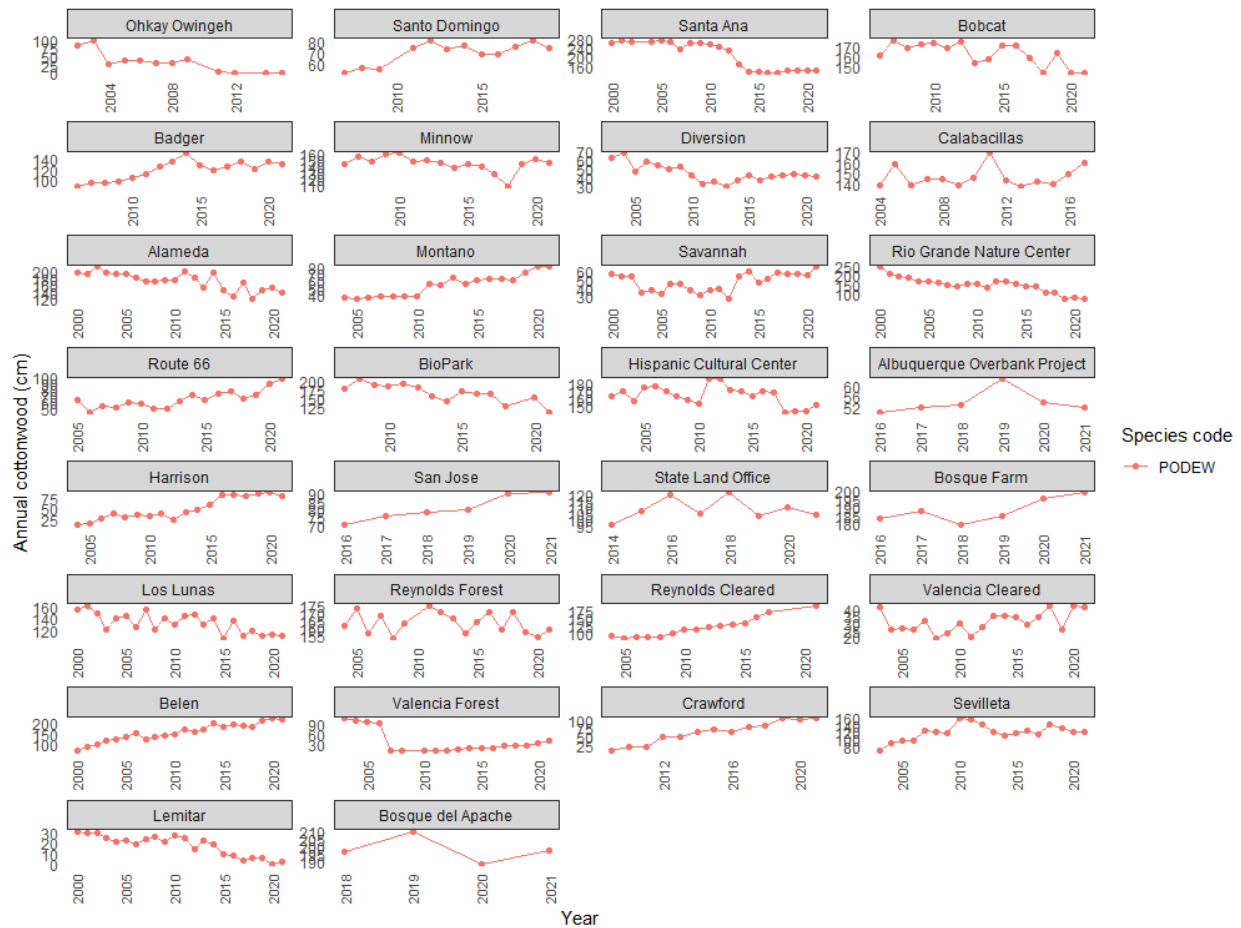


Figure 7.2. Cottonwood cover by site and year. The y-axis scale is independent for each site, so that trends are more apparent.

The slow decline of cottonwood is apparent in both the litterfall (leaf fall) and cover data (Figures 7.1 and 7.2). Spikes in leaf fall in recent years may represent a positive response (increasing productivity) to flood events; or, more specifically, to high groundwater and increasing variability in groundwater levels (see also Figures 5.10 and 5.11). This response can be delayed at flood sites, with decreased productivity during flood events and increased productivity the following year. Sites with younger trees, both pole plants or naturally-germinated saplings from overbank flooding show increasing cottonwood cover (Montano, Savannah, Route 66, AOP, Harrison, San Jose, Reynolds Cleared, Belen, and Crawford). Valencia Forest shows recovery of cottonwoods following the 2007 fire; these cottonwoods are reestablishing from root and shoot sprouts.



Figure 7.3 Monthly willow litterfall (g/m^2) shown across years at each site (listed north to south).

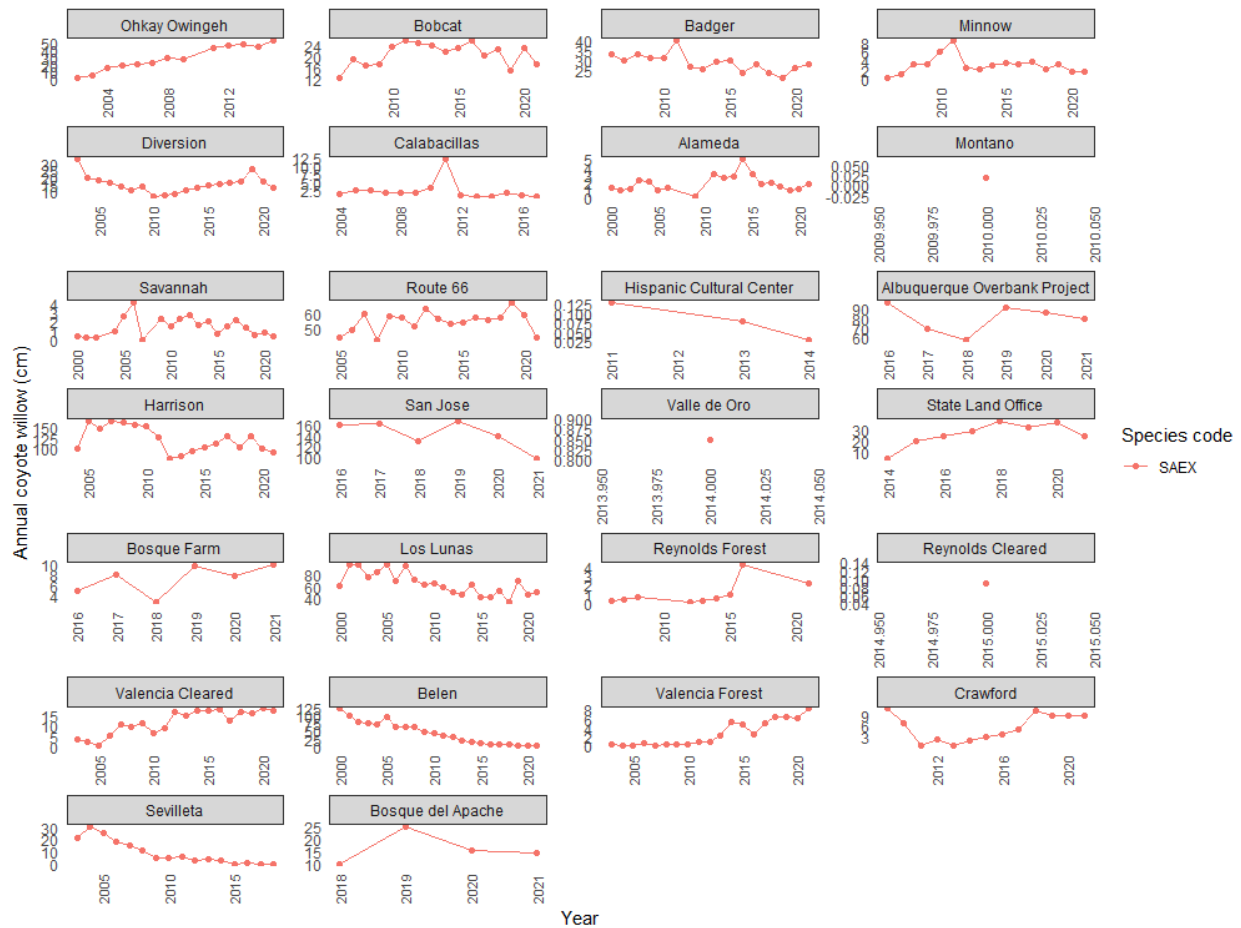


Figure 7.4. Coyote willow cover by site and year. The y-axis scale is independent for each site, so that trends are more apparent.

Willows, primarily coyote willow, show variability across sites with declines in sites with thicker, well-established vegetation (e.g., Los Lunas, Belen, Sevilleta) and increases at sites where there are openings for willow to expand (e.g., State Land Office, Reynolds Forest, Valencia Cleared, Valencia Forest, Crawford).

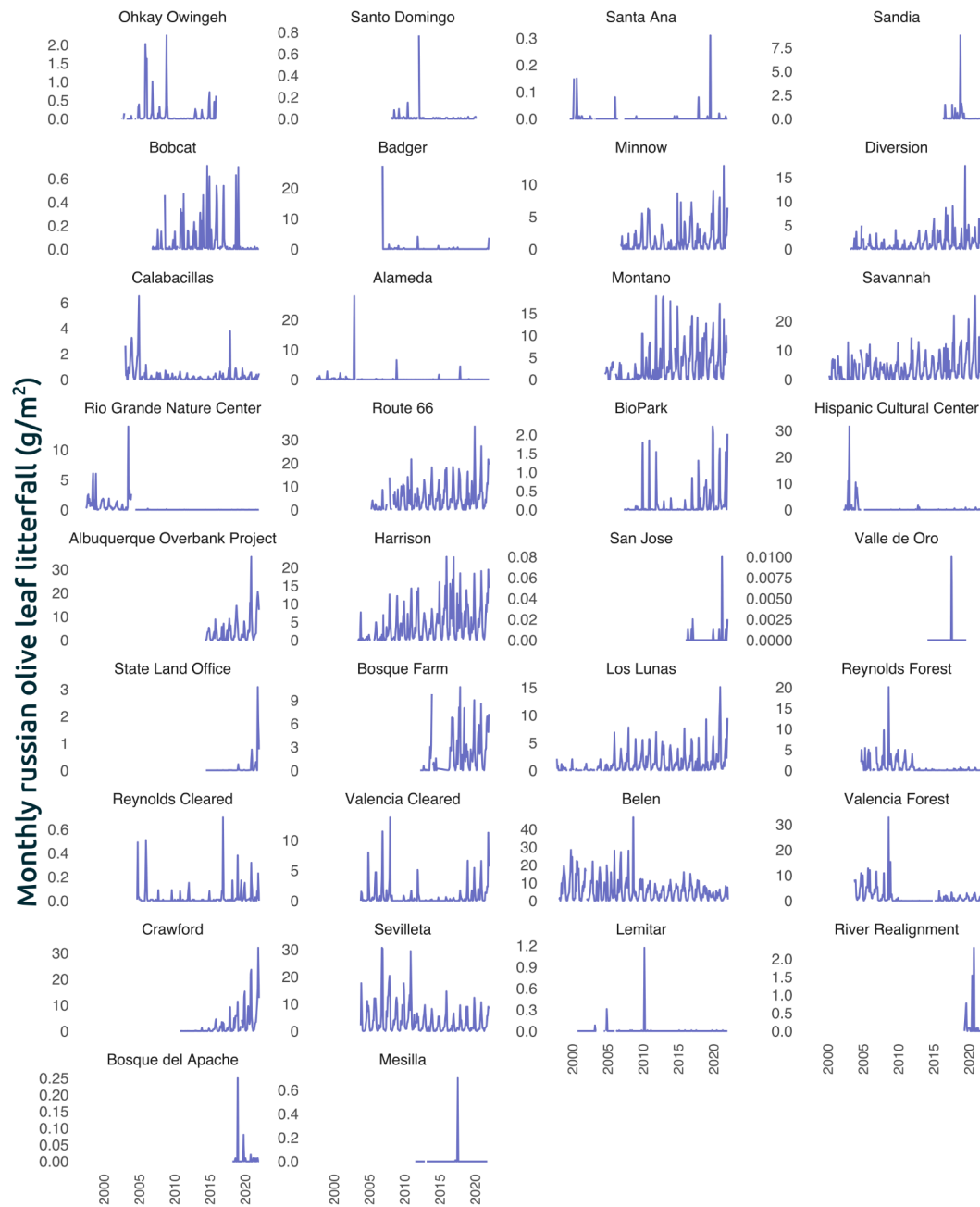


Figure 7.5 Monthly Russian olive litterfall (g/m²) shown across years at each site (listed north to south).

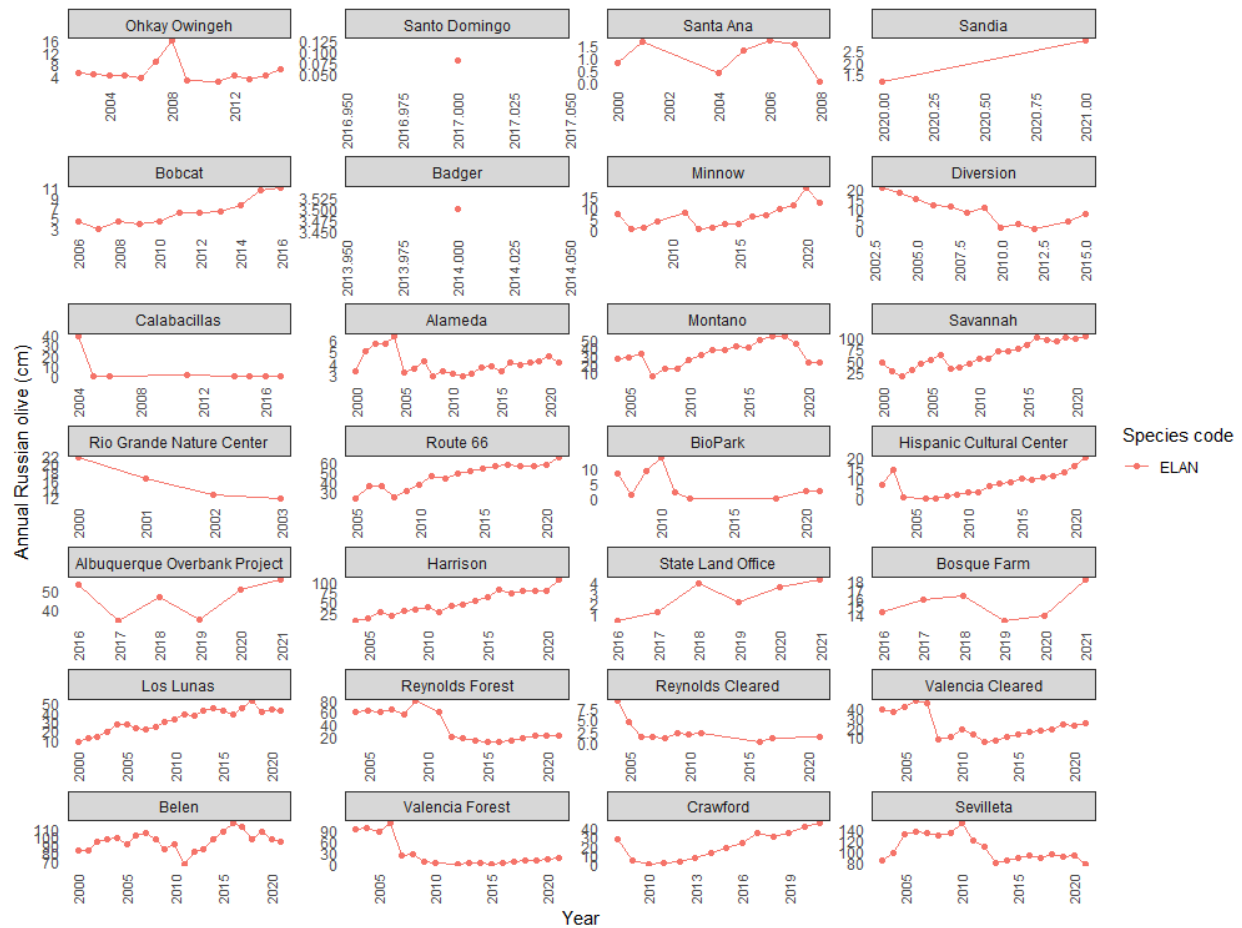


Figure 7.6. Russian olive cover by site and year. The y-axis scale is independent for each site, so that trends are more apparent.

Russian olive leaf production and cover remain steady with gradual increases at most sites, especially following clearing events. Even sites with thick stands are showing steady increases in cover (e.g., Savannah, Belen).

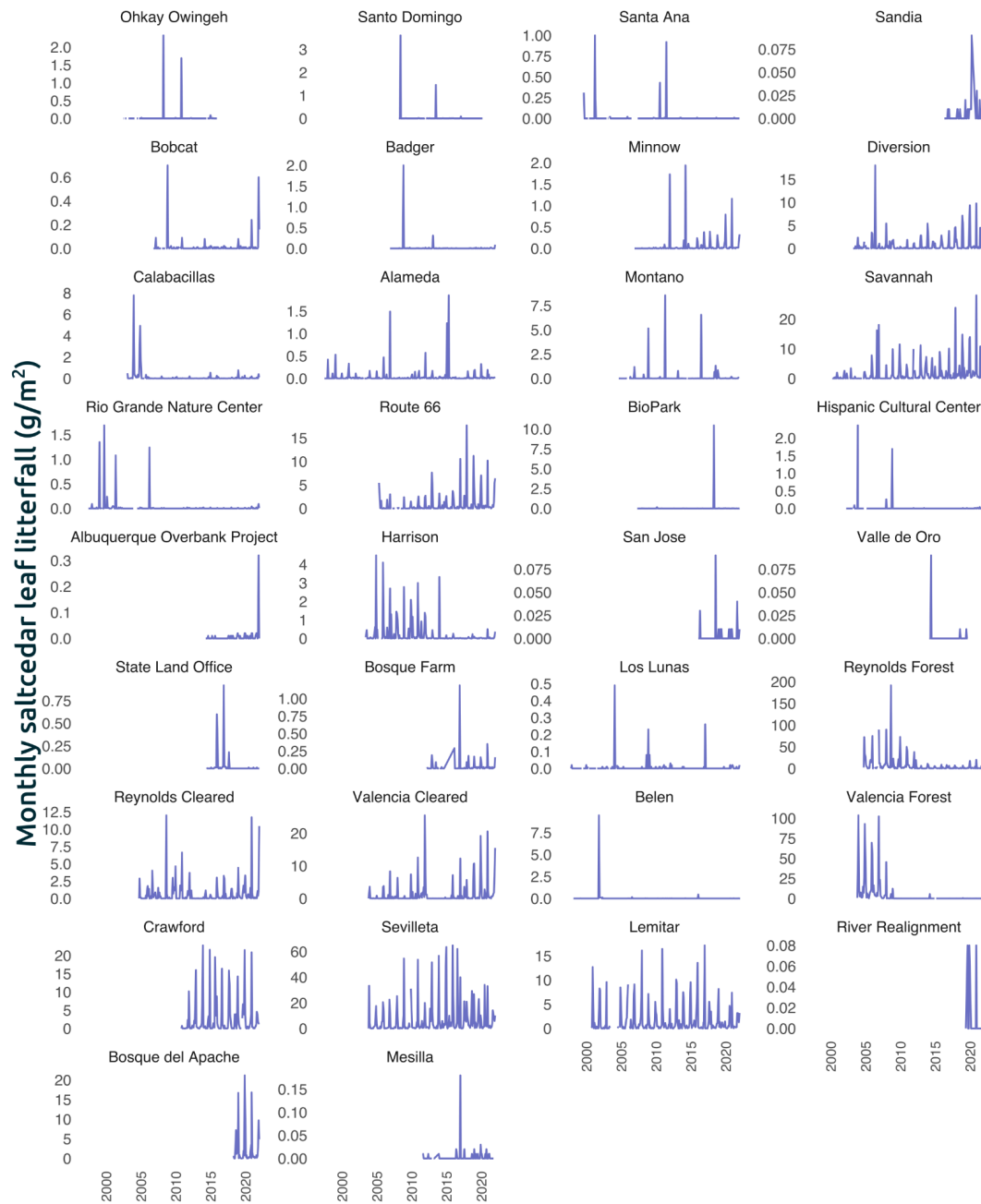


Figure 7.7 Monthly saltcedar litterfall (g/m²) shown across years at each site (listed north to south).

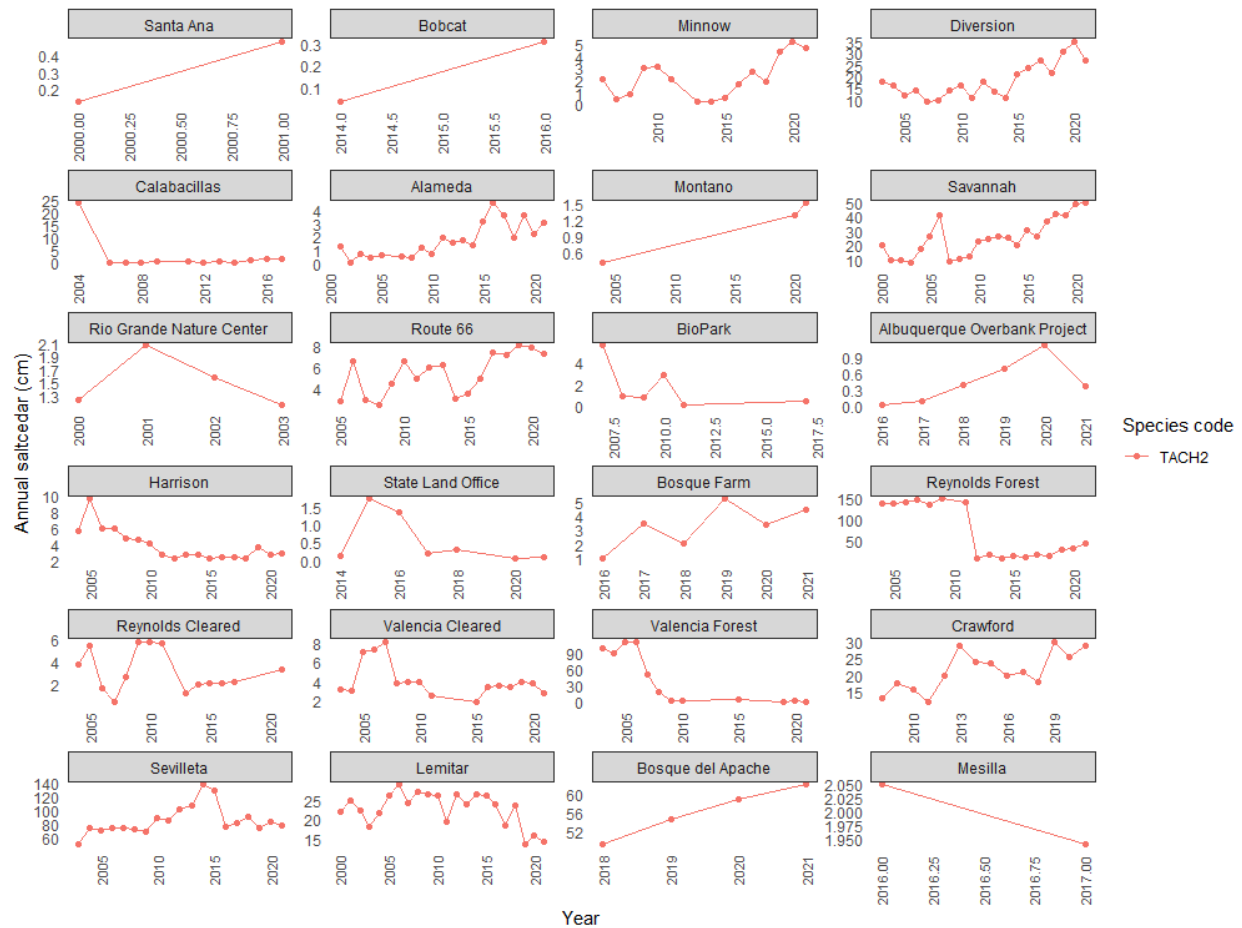


Figure 7.8. Saltcedar cover by site and year. The y-axis scale is independent for each site, so that trends are more apparent.

Saltcedar leaf production has shifted from one large peak in the fall to two peaks (one in summer due to defoliation by the tamarisk leaf beetle) and a second in the fall, resulting in lower monthly levels of leaf fall, but fairly steady annual levels (Figures 7.7 and 7.8). Saltcedar is increasing at more sites than not, although some sites do show declines in saltcedar (Harrison, Sevilleta, Lemitar) (Figure 7.8). Sudden drops in saltcedar cover are typically from clearing events or fire (e.g., Calabacillas, Reynolds Forest, Reynolds Cleared, Valencia Cleared, Valencia Forest) and are typically followed by gradual recovery.

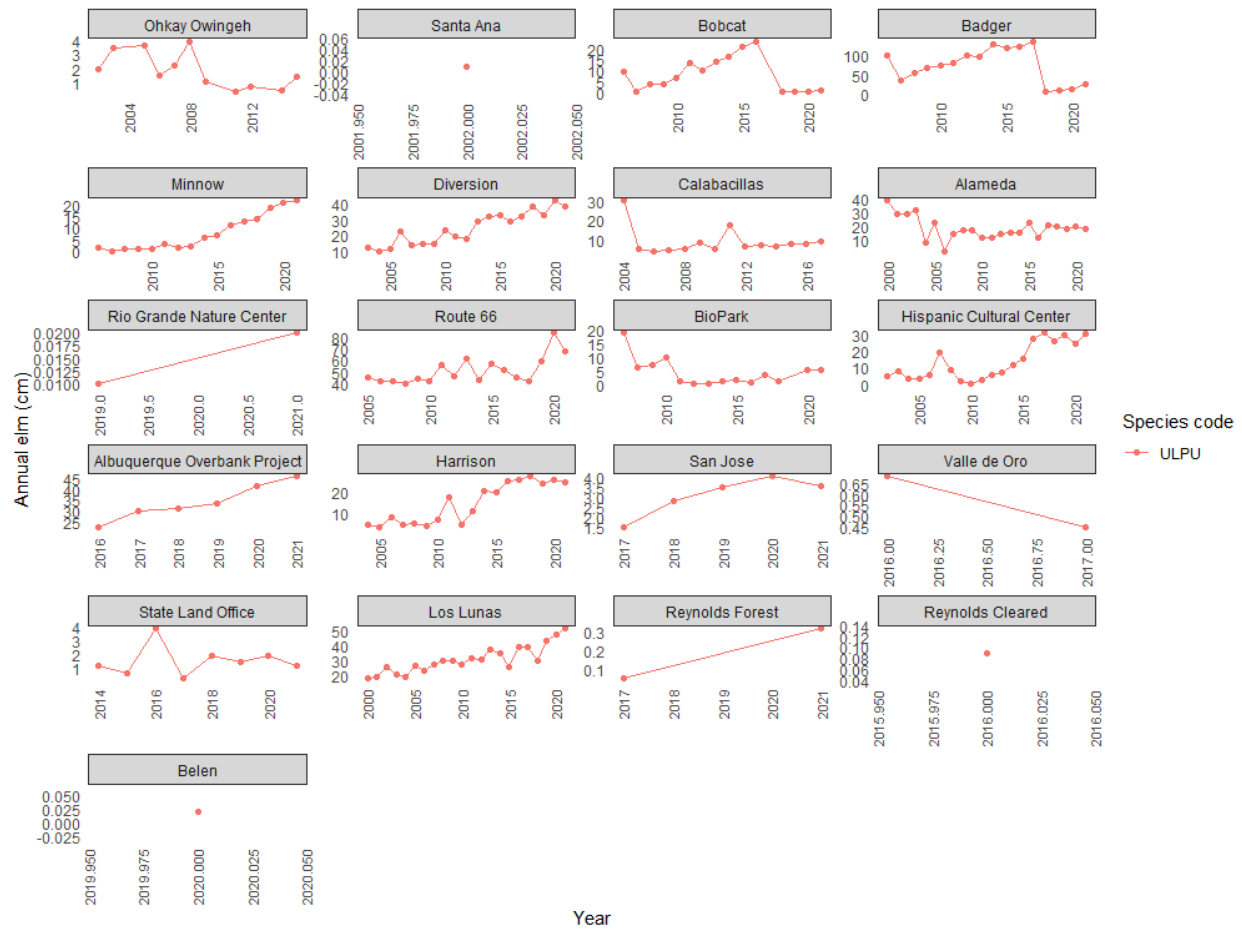


Figure 7.9. Siberian elm cover by site and year. The y-axis scale is independent for each site, so that trends are more apparent.

Elm cover is low where it is present at BEMP sites, but is increasing gradually at most of those sites.

Wetland plant species response to over time to changing depths to groundwater



Figure 7.10. Cover of dominant wetland species by site and year, for sites with at least one meter of wetland species cover. The y-axis scale is independent for each site, so that trends are more apparent. The species represented in this graph are: *Carex* sp., *Cyperus glomeratus* (exotic), *Cyperus* sp., *Equisetum laevigatum*, *Equisetum* sp., *Juncus arcticus*, *Juncus balticus*, *Juncus* sp., *Juncus torreyi*, *Lemna* sp., *Phragmites australis*, *Schoenoplectus* sp., *Schoenoplectus pungens*, *Schoenoplectus tabernaemontani*, *Typha latifolia*, and *Typha* sp.

Dominant wetland plants were analyzed by site and included sedges, rushes, horsetail, cattail, reeds, duckweed, and waterclover (Figure 7.10). The species represented in this analysis are: *Carex* sp., *Cyperus glomeratus* (exotic), *Cyperus* sp., *Equisetum laevigatum*, *Equisetum* sp., *Juncus arcticus*, *Juncus balticus*, *Juncus* sp., *Juncus torreyi*, *Lemna* sp., *Marsilea* sp., *Phragmites australis*, *Schoenoplectus americanus*, *Schoenoplectus* sp., *Scirpus* sp., *Scirpus pallidus*, *Schoenoplectus tabernaemontani*, *Schoenoplectus pungens*, *Schoenoplectus tabernaemontani*, *Typha latifolia*, and *Typha* sp. (not all species are shown on the graph). Wetland species, by

definition, rely on shallow water tables, although some are more resistant to groundwater fluctuations than others. *Carex* sp. cover is strongly tied to groundwater levels, with higher water tables supporting larger populations of *Carex*, although this sedge is rapidly declining (Los Lunas and Belen) or has disappeared from some BEMP sites (e.g., Route 66, Sevilleta) (Figure 7.10). *Equisetum* spp., or horsetail, cover is also strongly tied to changes in groundwater, with recovery seen in flood years. Rushes in the *Juncus* genus also fluctuate with changes in groundwater, with overall declines seen at some sites (Figure 7.10). Changes in precipitation do not correlate with changes in *Carex*, *Equisetum*, or *Schoenoplectus*. *Juncus* cover did show positive responses to precipitation events at Sevilleta until 2012, when overall declines in rushes outweighed any response to additional precipitation.

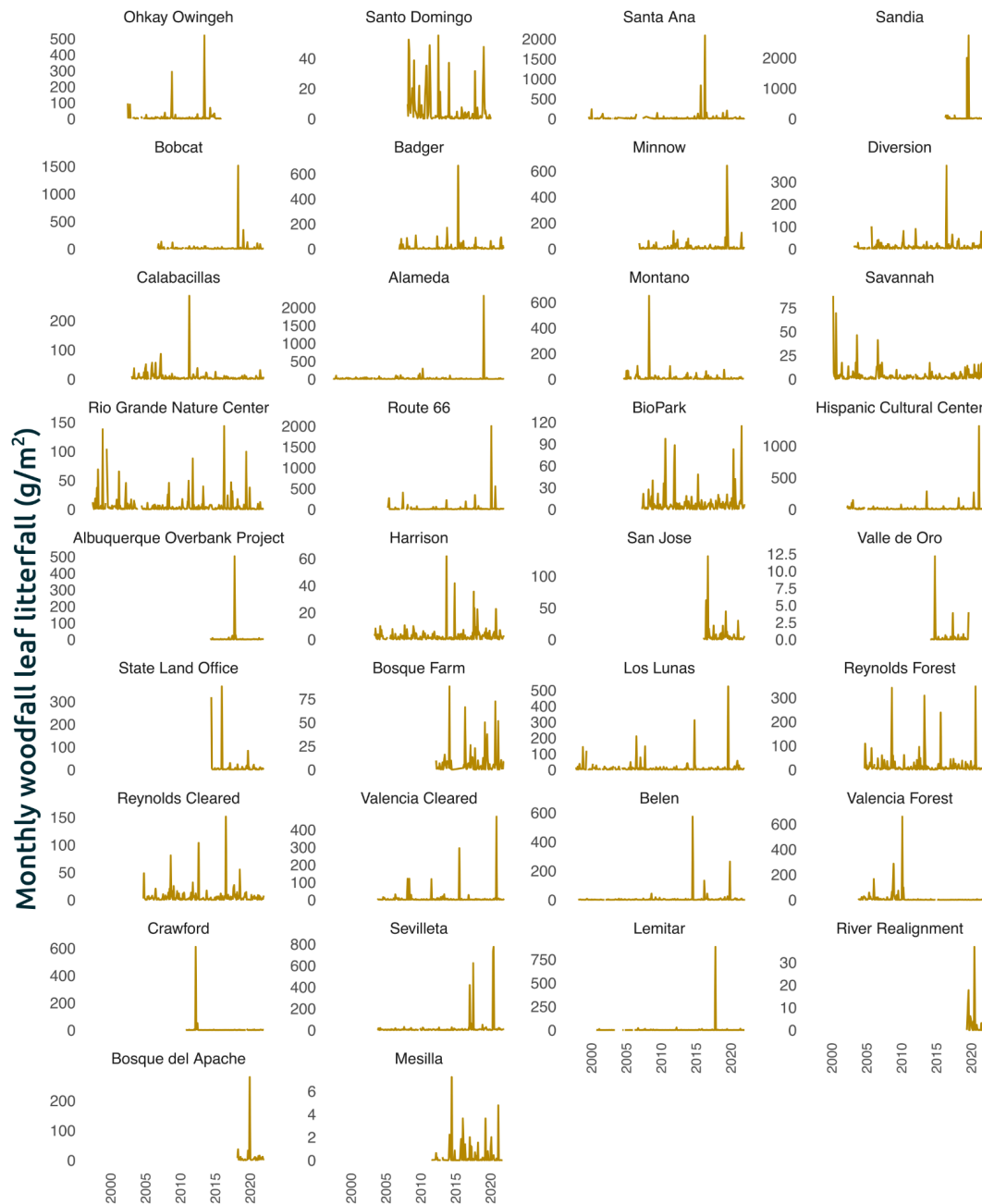


Figure 7.11. Monthly wood fall (g/m^2) shown across years at each site (listed north to south).

Wood fall is only captured through trees directly above the ten litterfall tubs dropping limbs into the tubs (as opposed to leaf fall, which is also influenced by winds). Even so, recent years show spikes in the twigs and branches falling at BEMP sites.

Post burn assessment



Figure 7.12. Rapid assessment of vegetation resprouts following the burn outside Bosque School in Albuquerque. At 19 days post-fire, Russian olive (RO), saltcedar (SC), and willow (WIL) were seen resprouting. There was also a small patch of yerba mansa (YM) resprouting. No cottonwoods had sprouted. By June 17, 2022 (23 days post-fire), New Mexico olive (NMO) was also resprouting.

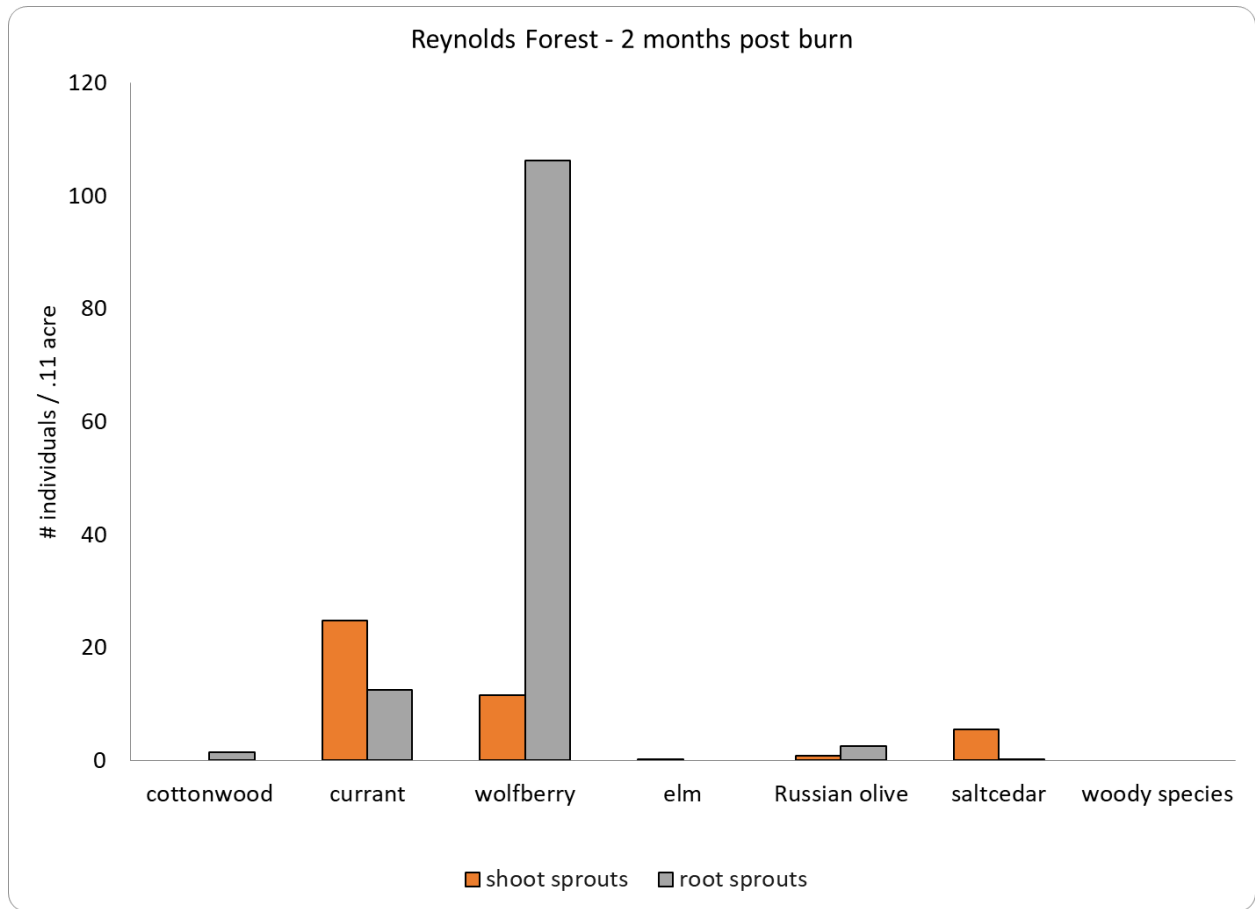


Figure 7.13. Shrub and tree sprouts at 2 months post-burn at the Reynolds Forest site in Belen (Big Hole fire).

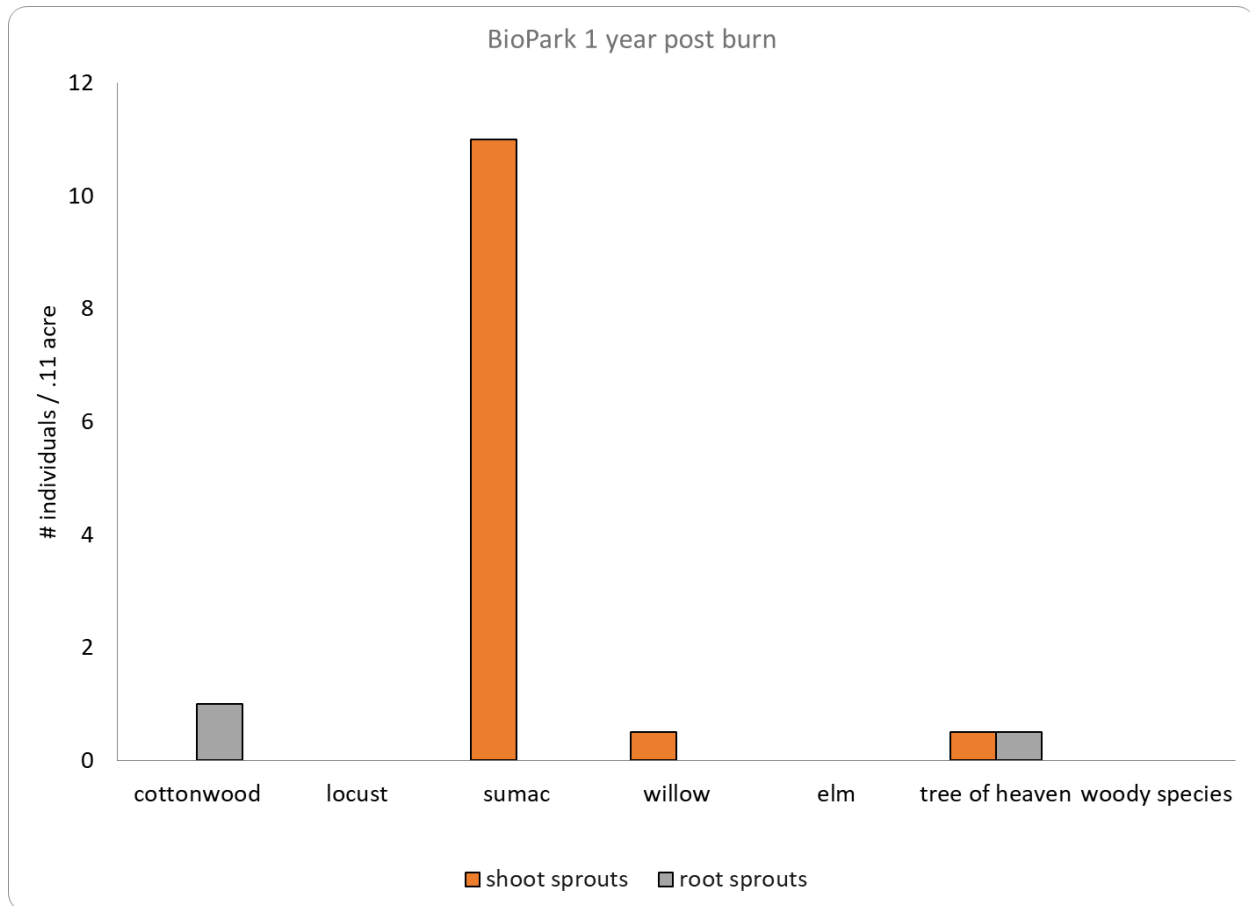


Figure 7.14. Shrub and tree sprouts one year post-burn at the BioPark site in Albuquerque.

The City of Albuquerque funded a Rapid Assessment UNM class pilot project in the summer of 2022. University of New Mexico students who had completed the Biology 408/508 course were able to sign up and participate in a class focused on in-depth field work and data summaries. During the summer, the students monitored post-burn sites; the Albuquerque fire east of Bosque School was a 2 week old fire, the Belen Big Hole fire was 2 months old, and the Albuquerque BioPark fire was a year old when the sites were monitored. Hink and Ohmart vegetation plots (30 x 15 m) were used to assess vegetation response. BEMP fuel load protocols were used to assess fuel load along four 50m transects. Ash depth and arthropods were also monitored.

Russian olive and saltcedar resprout faster than cottonwoods following fire, both from shoot sprouts and root sprouts (Figure 7.12). Cottonwoods have a low resprout rate and come back from root sprouts (Figures 7.13 and 7.14). Currant, wolfberry, and sumac are native shrubs that do well resprouting following fire (Figures 7.13 and 7.14). At the BioPark site, one year following the fire, the understory vegetation was dominated by kochia, tumbleweed, silverleaf

nightshade, and grasses. This was based on the four Hink and Ohmart vegetation plots. Further analyses will be conducted once all the 2022 vegetation transect data for that site are QA/QCed. We will also be able to closely track changes as the Reynolds Forest site recovers from the Big Hole fire.

8 Surface-Active Arthropods

Surface active arthropods are reported for 2020. 2021 and 2022 arthropod samples are being processed, entered and checked. 2023 collections are scheduled to begin May 2023 for all 33 BEMP sites.

Full monitoring methods can be found at:

https://github.com/BEMPscience/bemp_data/tree/master/surface_active_arthropods

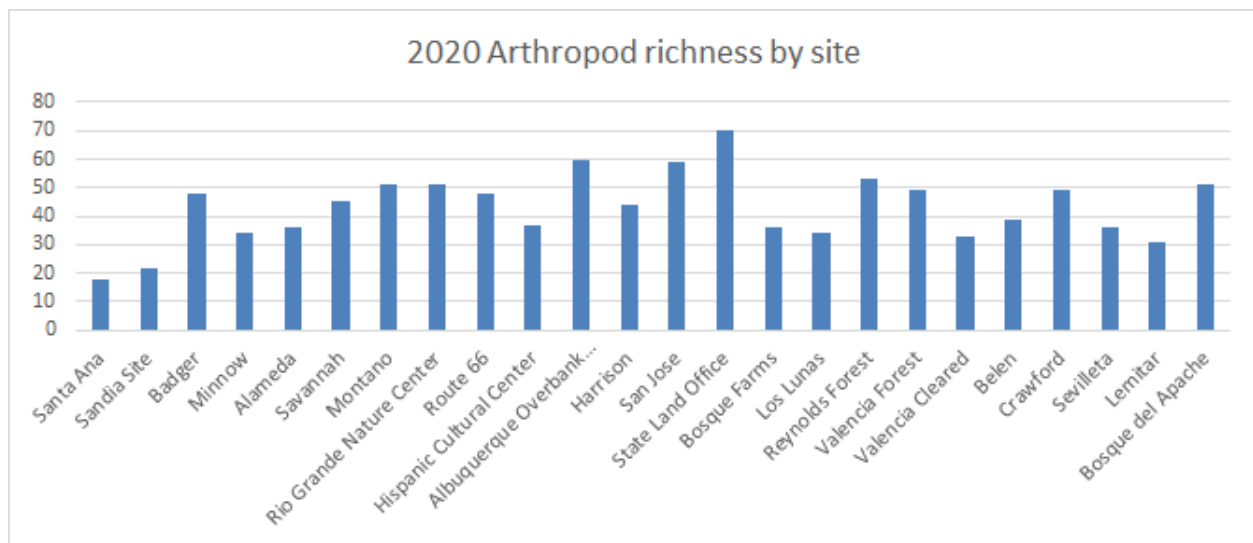


Figure 8.1. Arthropod richness for 2020. Sites arranged geographically north to south.

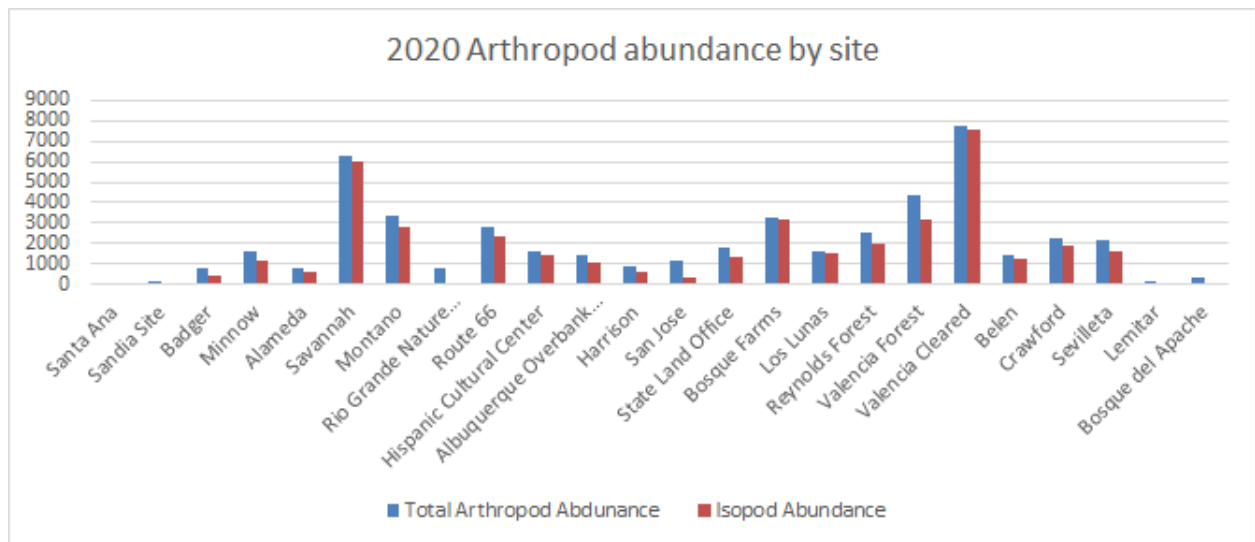


Figure 8.2. Isopod (red) and all arthropod abundance (blue) per site for 2020. Sites arranged geographically north to south.

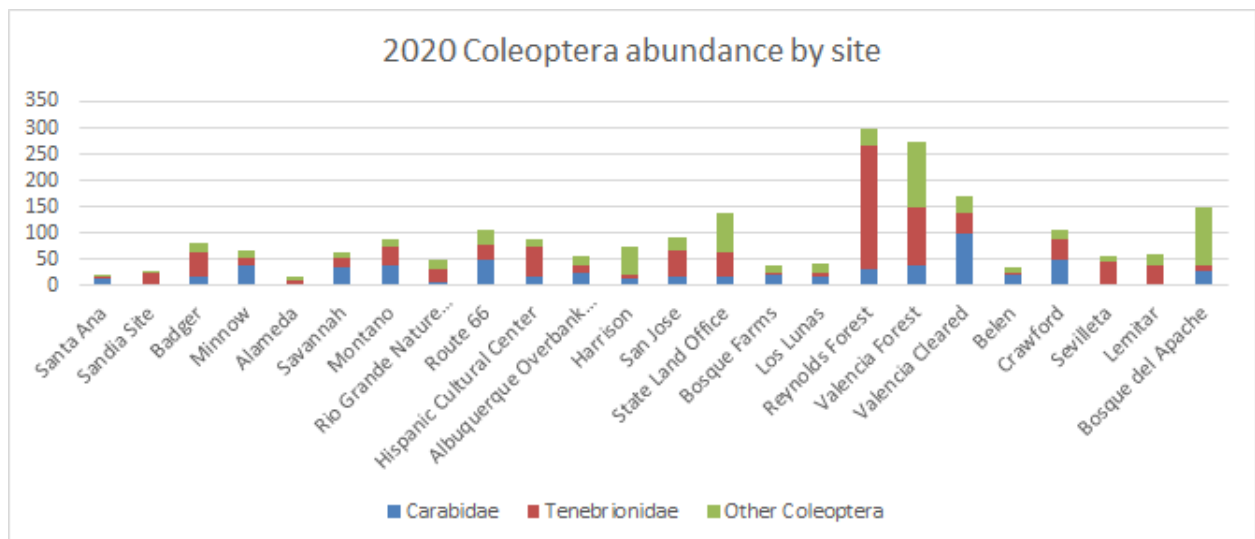


Figure 8.3. Coleoptera abundance per site for 2020, highlighting Carabidae (blue) and Tenebrionidae (red). Sites arranged geographically north to south.



Figure 8.4. Tenebrionidae and Carabidae mean annual sums over time across all BEMP sites arranged from north to south. These two families of beetles collected in the bosque are considered indicator species. Tenebrionidae are considered semi-arid while Carabidae prefer mesic or wetter habitats.

The vast majority of sites were dominated by isopods (both *Armadillidium vulgare* and *Porcelio laevis*) in 2020, as is typical in a non-high flow year (Figure 8.2). Even sites that experience regular flooding are dominated by isopods, where populations will drop during flood events. The driest sites (e.g., Santa Ana, Sandia, Lemitar) are not dominated by isopods.

Carabid and tenebrionid beetle abundance continue to reflect the general xeric to mesic to occasionally hydric environments at BEMP sites (Figures 8.3 and 8.4). Drier sites maintain dominant tenebrionid populations while dry sites that flood only during high flow events show the occasional spikes in carabid abundance. More mesic sites show greater fluctuations in both populations and sites that have more regular flooding and higher water tables support higher abundances of carabids.

Taxonomic assemblages as indicators of ecosystem

Latent Dirichlet Allocation (LDA) models are a class of unsupervised machine learning. The model is based on the idea that a sample site contains a collection of taxa (assemblages), you would expect taxa that are related to the sample site (and its environment) to appear more often in that site's data. Figure 8.5 and Table 8.1 show the model results for the San Jose BEMP site. From 2014 to 2016 we see diffuse assemblages of taxa that are related to drier environments. The flood in 2017 shifts these groups and from 2018 on there is a dominant group of taxa: *Dorymyrmex insanus*, *Tetramorium* sp., *Pogonomyrmex occidentalis*, *Crematogaster* sp., Acari, *Araeoschizus decipiens*. In figure 8.5 and Table 8.1, the assemblages for Bosque Farm are shown. Bosque Farms has a shallower depth to groundwater and we would expect taxa that are associated with mesic or wetter habitats. The dominant assemblage at Bosque Farms is: *Armadillidium vulgare*, *Calathus opaculus*, Curculionidae, *Porcelio laevis*, *Harpalus* sp., *Eleodes suturalis*, Lycosidae, Lepidoptera larva. During the 2017 flood at Bosque Farms we see a dip in the distribution of the assemblage which is due to the decrease of *Armadillidium vulgare*. It's worth noting that *Harpalus* sp. and *Eleodes suturalis* only show up in 2020 at Bosque Farms.

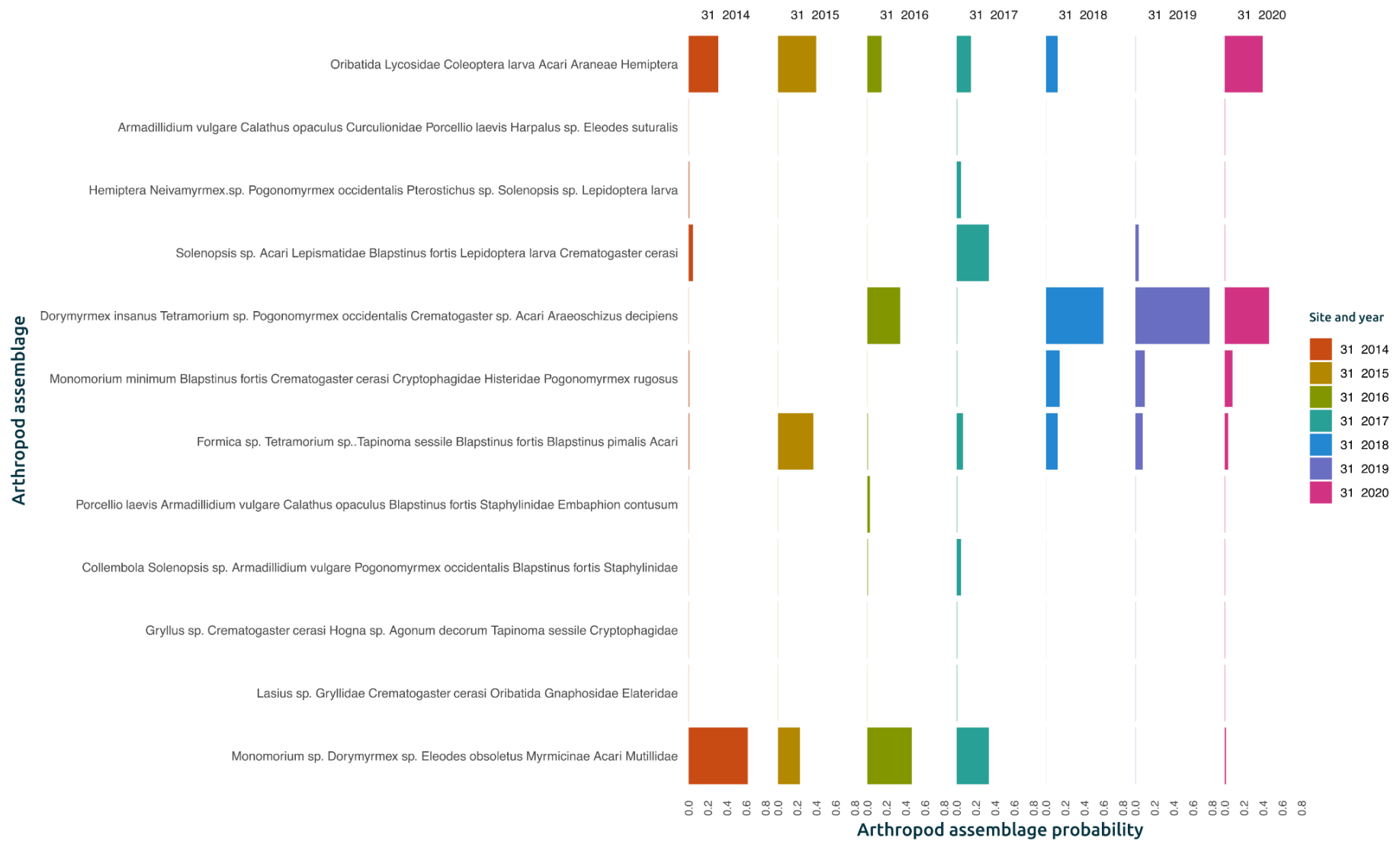


Figure 8.5. Latent Dirichlet Allocation (LDA) model of surface active arthropods at the San Jose site over time. The groups (top six taxa) on the left are the 12 groupings identified in the LDA model across all sites. The taxonomy is ordered from greatest abundance to least. During the flood year of 2017 and after we see a shift in the dominant arthropod groups.

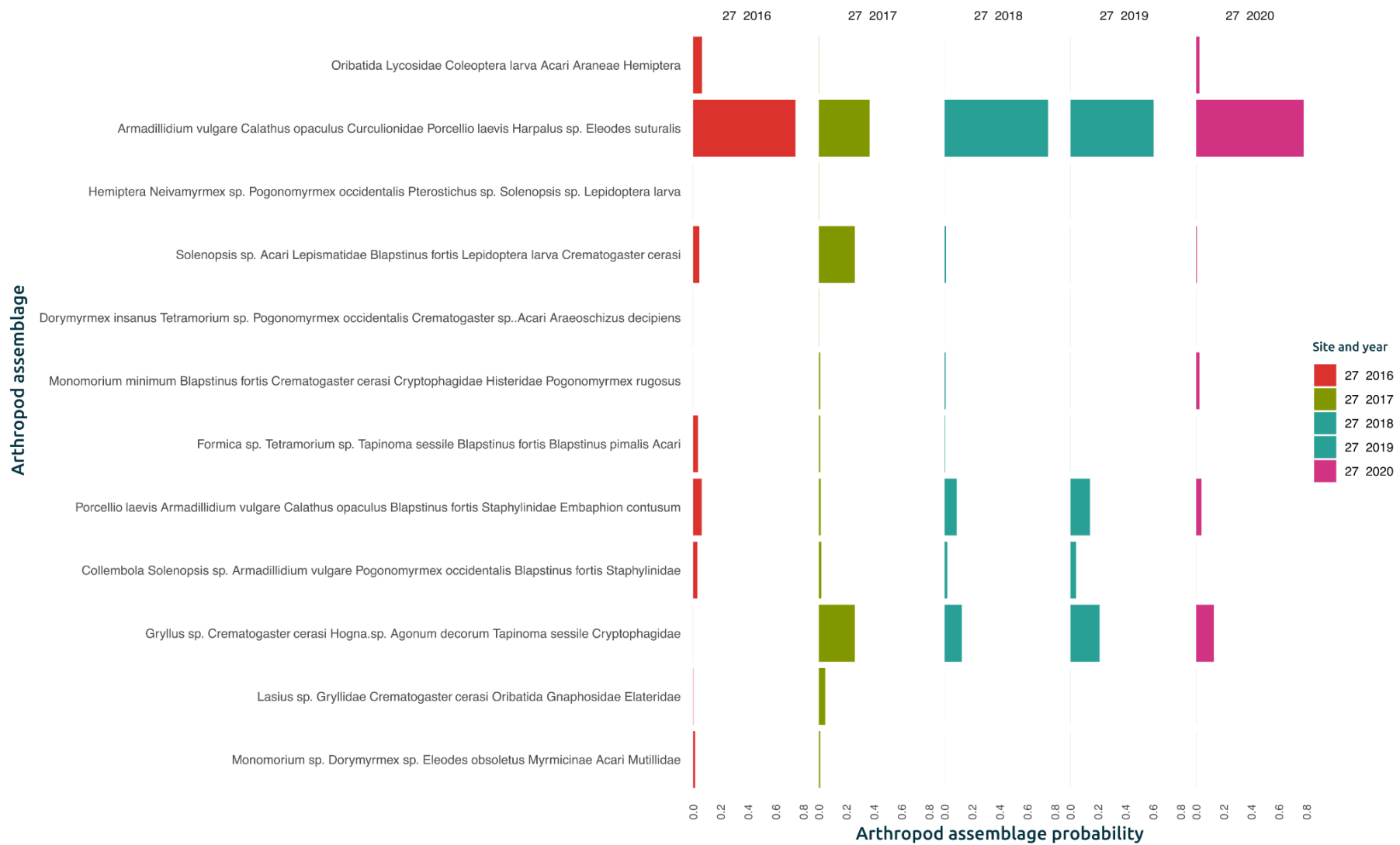


Figure 8.6. Latent Dirichlet Allocation (LDA) model of surface active arthropods at the Bosque Farms site over time. The groups (top six taxa) on the left are the 12 groupings identified in the LDA model across all sites. The taxonomy is ordered from greatest abundance to least. Bosque Farms is dominated by taxa that are associated with mesic or wetter habitats.

Table 8.1. LDA arthropod assemblages from Figures 8.5 and 8.6.

San Jose	Top assemblages
2014	<i>Monomorium</i> sp, <i>Dorymyrmex</i> sp, <i>Eleodes obsoletus</i> , Myrmicinae, Acari, Mutillidae
2015	1) Oribatida, Lycosidae, Coleoptera larva, Acari, Araneae, Hemiptera 2) <i>Formica</i> sp., <i>Tetramorium</i> sp., <i>Tapinoma sessile</i> , <i>Blapstinus fortis</i> , <i>Blapstinus pimalis</i> , Acari
2016	<i>Monomorium</i> sp., <i>Dorymyrmex</i> sp., <i>Eleodes obsoletus</i> , Myrmicinae, Acari, Mutillidae
2017	1) <i>Monomorium</i> sp <i>Dorymyrmex</i> sp., <i>Eleodes obsoletus</i> , Myrmicinae, Acari, Mutillidae 2) <i>Solenopsis</i> sp., Acari, Lepismatidae, <i>Blapstinus fortis</i> , Lepidoptera larva, <i>Crematogaster cerasi</i>
2018	<i>Dorymyrmex insanus</i> , <i>Tetramorium</i> sp., <i>Pogonomyrmex occidentalis</i> , <i>Crematogaster</i> sp., Acari, <i>Araeoschizus decipiens</i>
2019	<i>Dorymyrmex insanus</i> , <i>Tetramorium</i> sp., <i>Pogonomyrmex occidentalis</i> , <i>Crematogaster</i> sp., Acari, <i>Araeoschizus decipiens</i>
2020	1) <i>Dorymyrmex insanus</i> , <i>Tetramorium</i> sp., <i>Pogonomyrmex occidentalis</i> , <i>Crematogaster</i> sp., Acari, <i>Araeoschizus decipiens</i> 2) Oribatida, Lycosidae, Coleoptera larva, Acari, Araneae, Hemiptera
Bosque Farms	
2016 - 2020	<i>Armadillidium vulgare</i> , <i>Calathus opaculus</i> , Curculionidae, <i>Porcellio laevis</i> , <i>Harpalus</i> sp., <i>Eleodes suturalis</i> , Lycosidae, Lepidoptera larva

***Marinarozelotes barbatus* (L. Koch, 1866) update**

Marinarozelotes barbatus is an exotic spider species that was encountered in pit traps from the Albuquerque Overbank Project in 2016 and represented a new state record for this spider. In subsequent years this spider was found at several sites within Albuquerque, including San Jose, where it still maintains a significant presence. In 2020, a total of 100 individual spiders from five families and over 10 genera were collected at the San Jose site. Out of those, 55 were identified as *Marinarozelotes barbatus*, representing more than half of the spiders collected there.

In 2020 this spider was also collected at the Harrison, Albuquerque Overbank Project, and State Land Office sites. It was the first time that this spider was found at the State Land Office site, representing the southmost location where this spider has been documented in New Mexico.

9 Tamarisk Leaf Beetle

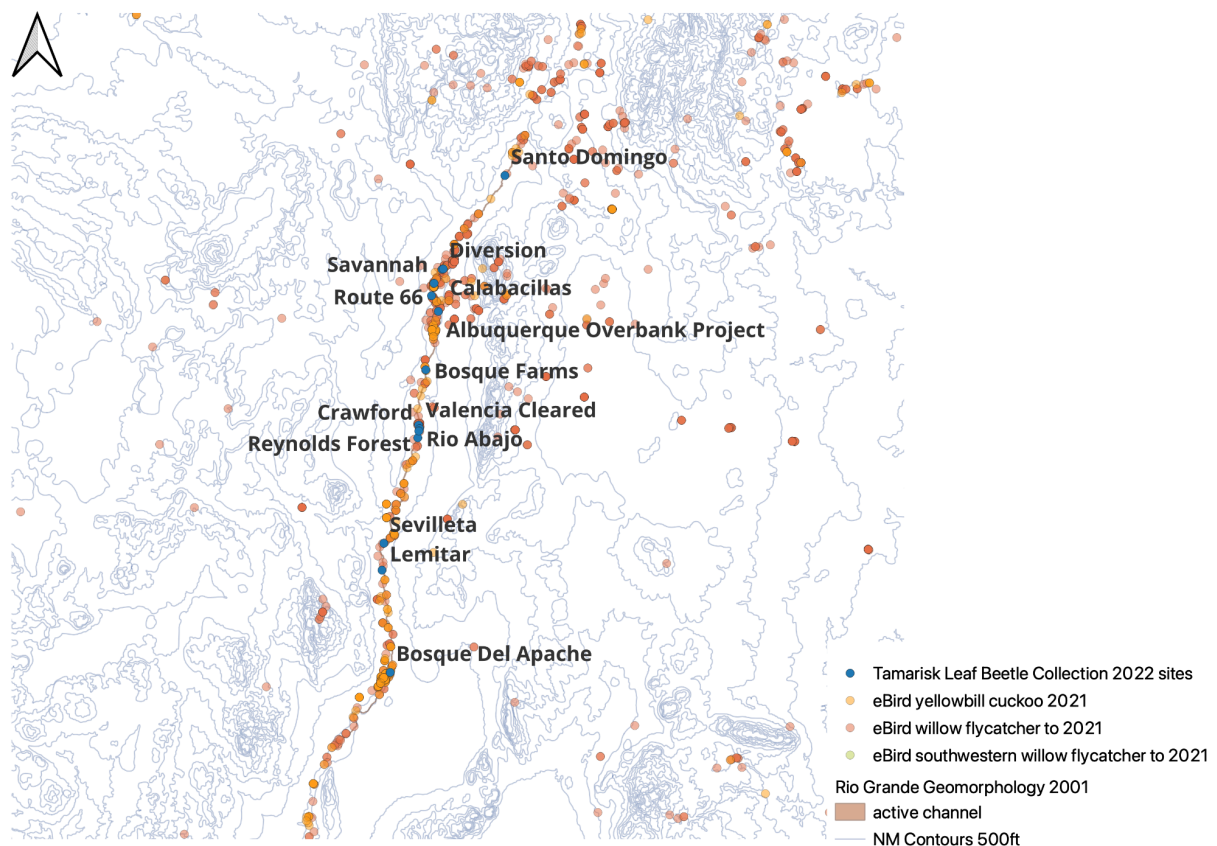


Figure 9.1. Current tamarisk leaf beetle monitoring sites for 2022 in bold. 500 foot contour intervals are in pale blue. Dark blue circles are TLB collection sites. Tan-brown circles are ebird data from 2021 that are species of interest.

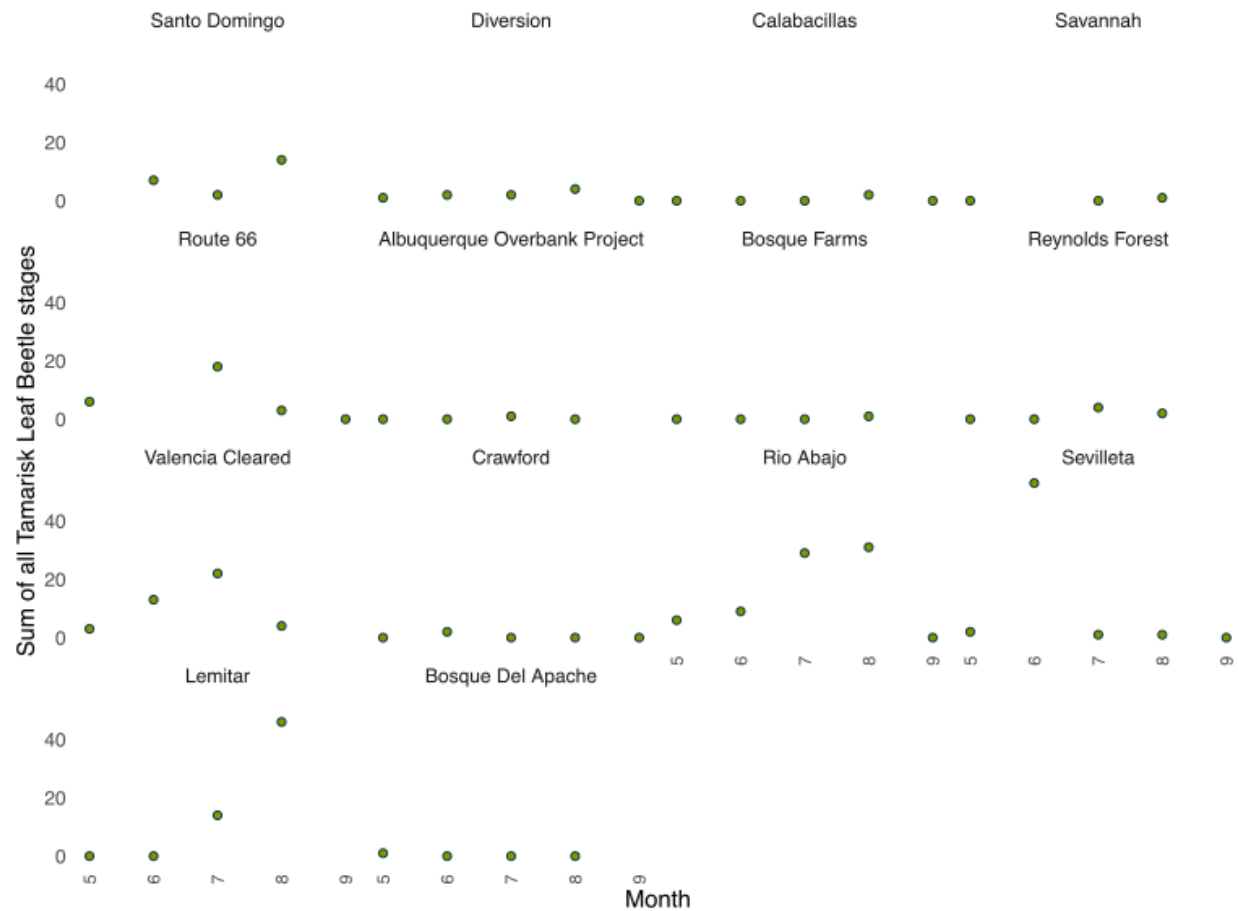


Figure 9.2. Annual (2022) sum of tamarisk leaf beetle abundance for all life stages. This includes egg masses, early and late larvae, and adults.

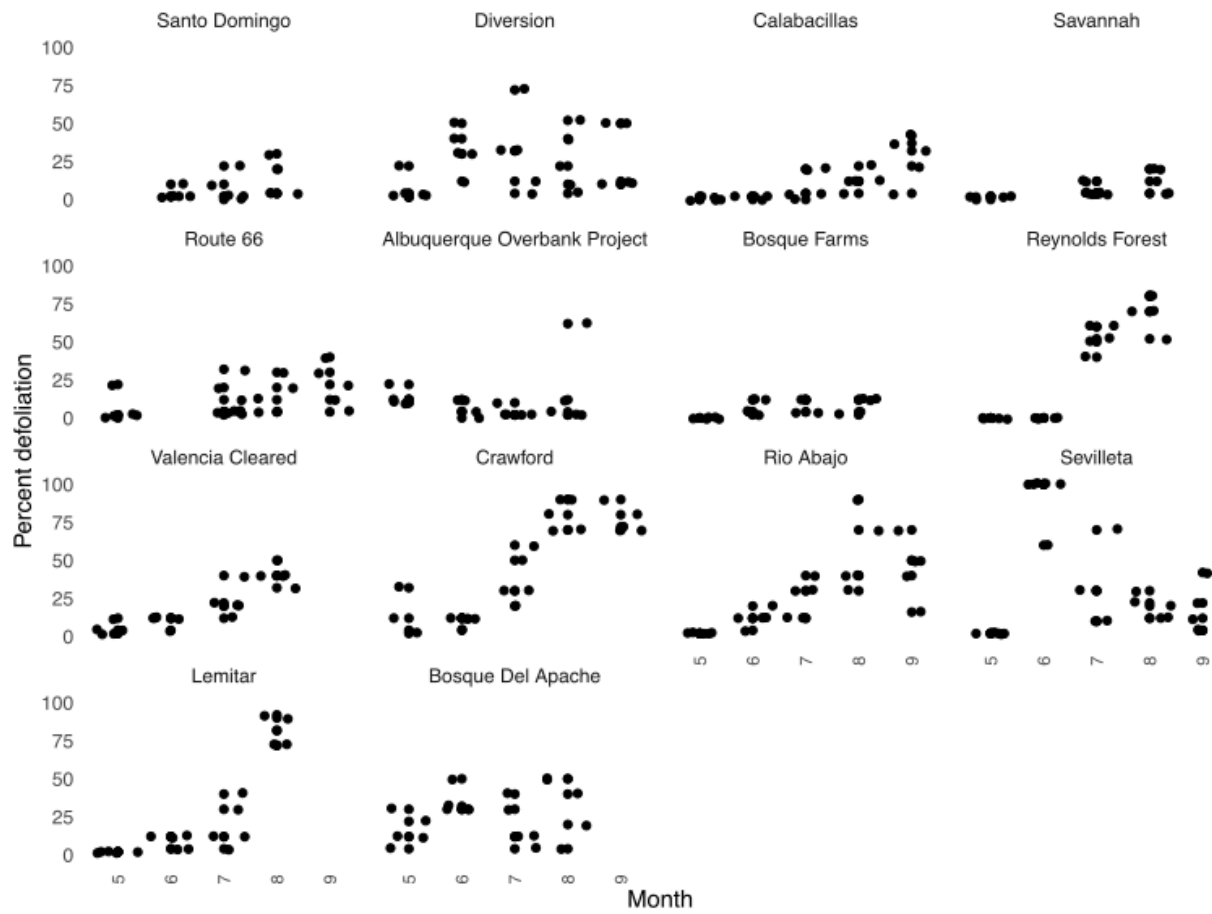


Figure 9.3. Percent total defoliation shown by tree across the sample sites for 2022. Total defoliation includes defoliation by both TLB and the leafhopper.

Table 9.1. Total TLB all life stages found at all sites May through September. Sites arranged from north to south.

Site name	May	June	July	August	September	Total
Santo Domingo	-	7	2	14	0	23
Calabacillas	0	0	0	2	0	2
Diversion	1	2	2	4	0	9
Savannah	0	0	0	1	0	1
Route 66	6	0	18	3	0	27
Albuquerque Overbank Project	0	0	1	0	0	1
Bosque Farms	0	0	0	1	0	1
Reynolds Forest	0	0	4	2	0	6
Valencia Cleared	3	13	22	4	0	42
Rio Abajo	6	9	29	31	0	75
Crawford	0	2	0	0	0	2
Lemitar	0	0	14	46	0	60
Sevilleta	2	53	1	1	0	57
Bosque Del Apache	1	0	0	0	0	1

Tamarisk leaf beetles were seen and collected at all 14 sites (including the one site funded by the Greater Rio Grande Water Alliance) (Figure 9.1) . The abundance varied from site to site and ranged from one TLB collected at the Albuquerque Overbank Project, Bosque del Apache, Bosque Farms and Savannah sites to 75 TLBs collected at the Rio Abajo site over the entire collection season (Figures 9.2-9.3, Table 8.1).

The highest quantity of TLBs for all sites were collected in July with 104 individuals collected, followed closely in August with 95 individuals collected. The individual sites followed this trend with the exception of Sevilleta which had peak abundance in June, and Lemitar which had peak abundance in August. Rio Abajo had a relatively similar abundance in July and August with 29, and 31 individuals collected respectively for those months. No TLBs were collected at the six sites sampled in September (Diversion, Route 66, Rio Abajo, Crawford, and

Sevilleta). This is similar to previous years and may indicate a lack of TLB activity in these areas after August.

Despite the varied TLB numbers, defoliation levels ranged from 0 to 100% at the sites (Figure 9.3). Defoliation due to the tamarisk leafhopper, which results in yellow foliage, ranged from 0 to 70%, while defoliation due to the TLB, which results in brown/dead foliage, ranged from 0 to 100%, as was seen in total defoliation (which is both yellow + brown defoliation) (Figure 9.3). The high defoliation levels at some sites underline the importance of the full range of monitoring, as the beetle populations were likely missed on the specific sampling date, but the evidence of beetle damage was still observable and quantifiable.

10 Outreach

During the 2021-22 school year, BEMP's operations were still impacted by the ongoing COVID-19 pandemic. BEMP educators continued to deliver in person and virtual programs, as well as offering activities and handouts through our online and social media platforms. BEMP reached 4095 students and 851 adults in person, both in the field and in the classroom. Field education occurred through monitoring collections (monthly monitoring, surface-active arthropod trapping, and water quality monitoring) as well as through study trips (primarily focused on stormwater science, phenology, and nature journaling). We continued to provide educational programming through online printable and electronic platforms, including self-led printable activities and video lessons. BEMP reached 11,920 participants through our handouts. Undergraduate and graduate students participating in the Biology 408/508 course at UNM conducted field and lab work during this time; and, an in-depth research course was provided in the summer and fall for UNM students. In the summer of 2022, BEMP supported the Horizons program, Latino Outdoors, and Ironwood Tree Experience.

Throughout August 2021-May 2022, BEMP education collaborated with 29 different schools (up from 9 the previous year), including: A. Montoya Elementary School, ACE Leadership High School, Bandelier Elementary School, Bosque School, Central New Mexico Community College, College and Career High School, Coyote Willow Family School, East Mountain High School, East San Jose Elementary School, Holy Ghost Academy, Hot Springs High School, Inez Elementary School, Jefferson Middle School, La Academia de Esperanza Charter School, La Cueva High School, Lacordaire Academy, Magic City Acceptance Academy, Mountain Mahogany Community School, Native American Community Academy, Natures Ninos, Nex+Gen Academy, North Star Elementary School, Rio Rancho Cyber Academy, Rudolfo Anaya Elementary School, South Valley Academy, St. Mary's School, The International School at Mesa del Sol, Valle Vista Elementary School, and Wilson Middle School. The majority of these schools are classified as Title I schools, wherein at least 40% of students qualify for free and/or reduced lunch.

BEMP educational materials are NGSS-aligned and engage students in real data analysis within their own backyards. These materials are available on our website and through our publicly accessible GitHub page to ensure broad accessibility. All educational activities and materials are

provided bilingually in Spanish and English, strengthening accessibility initiatives for broader audiences. BEMP has maintained a social media presence across Instagram, Facebook, and YouTube's online platforms.

11 Implications for management

Shifts in water availability, from long-term declining river flows and declining groundwater levels, to changes in the timing of precipitation and increasing variability in the system, will have profound impacts on the plant and animal communities of the Rio Grande. Already, changes in native wetland vegetation cover can be seen, as these plants are more responsive to groundwater availability than precipitation inputs. Monsoon rains and late tropical storms do increase river flows, but often those flows move too quickly to allow groundwater response, so species dependent on groundwater (phreatophytes) are more likely to be negatively impacted as the hydrograph changes. Arthropod assemblages also respond to shifts in water availability, both directly and through responsiveness to plant community changes. Arthropods are a straightforward way of tracking impacts to fauna and tracking shifts in the environment.